Supplementary material for "Holomorphic Feynman diagrams in four dimensions and the quantum BPS ring": algorithm for the lifting tests

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This document is the supplementary material of the article [1]. Sections, equations and statements of this supplement carry the prefix S; references without that prefix are to the article.

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We compute the lifting tests of Section 9 up to a fixed level by reducing each charge sector to finite-dimensional linear algebra.

The algorithm below produces the cohomology counts and obstruction ranks; the later sections apply it theory by theory and give explicit representatives and their corrections.

Its outputs are $h = \dim E_1$ with $E_1 = H(Q_0)$, the Cartan-restriction rank ρ, the fortuitous count f, and the ranks of the successive differentials d_r.

The computed sector tables are in quantum-lifts-results.

S1 The census algorithm

We construct the cochains in each charge sector, evaluate the loop corrections, and compute the successive obstruction ranks.

Cartan restriction and comparison with stable cohomology then identify the non-Cartan and fortuitous classes, while exact and finite-field checks record how the results are certified.

S1.1 Charge sectors and the grading of the loop corrections

For $\mathcal{N} = 4$ Yang–Mills theory, we organize the cochains by their letter weights and derivative counts.

Counting the vertices and contractions gives the grading of each loop correction, which determines its target sector and bounds the possible obstruction orders.

For the $\mathcal{N} = 4$ letters, θ_I counts the Ith odd-coordinate weight in C, and $n_\pm$ counts $\partial_{+\pm}$ derivatives.

For a monomial with Y letters, let e_I be the Ith unit vector and write

$$\begin{aligned} &\lambda = (\theta_1, \theta_2, \theta_3; n_+, n_-), \qquad \Theta = \theta_1 + \theta_2 + \theta_3, \qquad \mathsf{d} = (1,1,1;-1,-1), \\ &\theta(\gamma^I) = e_I, \qquad \theta(\beta_I) = (1,1,1) - e_I, \qquad \theta(\mathsf{b}) = (1,1,1), \qquad \theta(\mathsf{c}) = 0, \end{aligned} \tag{S1.1}$$

$C^Y(\lambda)$, also written $(\lambda)_Y$, is the cochain sector with weight λ and Y letters. Its component degree is $\Theta = \deg_\theta$.

$$\deg_\theta \mathsf{c} = 0, \quad \deg_\theta \gamma = 1, \quad \deg_\theta \beta = 2, \quad \deg_\theta \mathsf{b} = 3, \qquad \deg_\theta(\partial_{+\dot a} X) = \deg_\theta X, \tag{S1.2}$$

At order n, a product graph has n cubic vertices and $2n$ physical edges, each pairing component degrees that sum to three.

$$\deg_\theta \mathcal{O}_n = \deg_\theta \mathcal{O}_0 + 3n - 6n = \deg_\theta \mathcal{O}_0 - 3n, \qquad \mathcal{O}_n = 0 \quad \text{if } 3n > \deg_\theta \mathcal{O}_0. \tag{S1.3}$$

The degree losses add under iterated products.

For a nonzero $Q_n X$, the $n+1$ cubic vertices and $2n+1$ edges give

$$\deg_\theta(Q_n X) = \deg_\theta X + 3(n+1) - 3(2n+1) = \deg_\theta X - 3n \tag{S1.4}$$

The physical charges are the dimension E, the three $SU(4)_R$ Cartan charges R_I, and the spins J_L, J_R.

$$\begin{aligned} E &= \tfrac{1}{2}(\Theta + Y) + n_+ + n_-, \qquad R_I = \theta_I - \tfrac{1}{2}(\Theta - Y), \\ J_L &= \tfrac{1}{2}(\Theta - Y + n_+ + n_-), \qquad J_R = \tfrac{1}{2}(n_+ - n_-), \\ E &= R_1 + R_2 + R_3 + 2J_L, \qquad 2(E + J_L) = 2\Theta + 3(n_+ + n_-). \end{aligned} \tag{S1.5}$$

letter	physical field	θ	(n_+, n_-)	E	(R_1, R_2, R_3)	J_L	J_R
γ^I	scalar $\bar\phi_I$	e_I	$(0,0)$	1	e_I	0	0
β_I	fermion ψ^I_+	$(1,1,1) - e_I$	$(0,0)$	$\frac{3}{2}$	$(\frac{1}{2},\frac{1}{2},\frac{1}{2}) - e_I$	$\frac{1}{2}$	0
b	field strength f_{++}	$(1,1,1)$	$(0,0)$	2	$(0,0,0)$	1	0
$\partial_{+\pm}\mathsf{c}$	gaugino $\bar\lambda_\pm$	0	$(1,0),(0,1)$	$\frac{3}{2}$	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	0	$\pm\frac{1}{2}$
$\partial_{+\pm}$	covariant derivative $\nabla_{+\pm}$	0	$(1,0),(0,1)$	1	$(0,0,0)$	$\frac{1}{2}$	$\pm\frac{1}{2}$

The level is $E + J_L$. Equation (S1.5) makes its set of sectors finite.

Example Threshold charges

The $SU(2)$ threshold class (S8.25) has two γ, five β and zero derivatives.

$$\lambda = (4,4,4;0,0), \quad Y = 7,$$
$$E = \frac{19}{2}, \qquad R_I = \frac{3}{2}, \qquad J_L = \frac{5}{2}, \qquad E + J_L = 12.$$

The current form (B.19) has degree n in each dotted component, giving

$$\mathcal{I}_\Gamma^{\text{raw cut}}(\lambda_{v,+\dot a} \mapsto c\lambda_{v,+\dot a};\ z_e^{+\dot a} \mapsto c^{-1} z_e^{+\dot a}) = c^n\, \mathcal{I}_\Gamma^{\text{raw cut}}(\lambda; z), \qquad c > 0. \tag{S1.6}$$

The λ powers supply output derivatives and the z powers remove input derivatives. Hence Q_n adds n derivatives in each direction [2, Equations (3.44)–(3.45)].

Q_0 preserves both totals. Choose homogeneous contractions (i_r, π_r, h_r) in (8.23).

Proposition S1.1.

For $n \geq 0$ and $r \geq 1$,

$$Q_n : C^Y(\lambda) \to C^{Y+1-n}(\lambda - n\mathsf{d}), \qquad d_r : E_r^Y(\lambda) \to E_r^{Y+1-r}(\lambda - r\mathsf{d}). \tag{S1.7}$$

Both maps have $(\Delta E; \Delta R_I; \Delta J_L, \Delta J_R) = (\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, 0)$.

Proof

$$\Delta Y = 3(n+1) - 2(2n+1) = 1 - n, \qquad \theta_I(Q_n X) - \theta_I(X) = (n+1) - (2n+1) = -n.$$

The vertex and edge counts give the letter and flavour shifts. Equation (S1.6) supplies n derivatives in each direction.

The homogeneous contractions and transferred coefficients obey

$$\begin{aligned} h_r &: E_r^Y(\lambda) \longrightarrow E_r^{Y+r-1}(\lambda + r\mathsf{d}), \\ [\hbar^m]\mathbf{Q}^{(r)} &: E_r^Y(\lambda) \longrightarrow E_r^{Y+1-r-m}(\lambda - (r+m)\mathsf{d}). \end{aligned} \tag{S1.8}$$

For $[\hbar^m]\mathbf{Q}^{(r+1)}$, let ℓ count perturbation factors. Their orders sum to $m+1$, and the term contains $\ell - 1$ copies of h_r.

$$\begin{aligned} \Delta Y &= \ell(1-r) - (m+1) + (\ell-1)(r-1) = -r - m, \\ \Delta\lambda &= -(\ell r + m + 1)\mathsf{d} + (\ell - 1)r\mathsf{d} = -(r+m+1)\mathsf{d}. \end{aligned} \tag{S1.9}$$

This proves the induction from $\mathbf{Q}^{(0)} = \mathbf{Q}$. Set $m = 0$ for d_{r+1} and insert the shifts into (S1.5).

$$\Delta\Theta = -3n, \qquad \Delta(n_+ + n_-) = 2n, \qquad (\Delta E; \Delta R_I; \Delta J_L, \Delta J_R) = \left(\frac{1}{2}; \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; -\frac{1}{2}, 0\right).$$

□

The sectors $(\lambda + k\mathsf{d})_{Y+k}$, $k \in \mathbb{Z}$, have the same physical charges. The cochain grading distinguishes the loop orders.

Corollary S1.2.

An outgoing d_r requires $r \leq \min_I \theta_I$, and an incoming d_r requires $r \leq \min(n_+, n_-)$.

A class with zero input derivatives has no incoming d_r for $r \geq 1$.

Proof

$$\theta_I - r \geq 0 \iff r \leq \min_I \theta_I, \qquad n_\pm - r \geq 0 \iff r \leq \min(n_+, n_-).$$

□

S1.2 Highest-weight cochains

We reduce the $\mathcal{N} = 4$ calculation to highest-weight cochains so that each symmetry multiplet is counted once.

We construct spanning sets from letter and derivative data, then compare their ranks with the exact dimensions obtained from group characters.

The flavour raising operators E_{12}, E_{23} and the dotted-index raising operator J_+ commute with Q_n by the $SL(3)$ invariance and spinor contractions in (7.3).

Compute the maps on their common highest-weight space.

$$C^Y_{\mathrm{HW}}(\lambda) = C^Y(\lambda) \cap \ker E_{12} \cap \ker E_{23} \cap \ker J_+, \qquad \theta_1 \geq \theta_2 \geq \theta_3 \geq 0, \qquad n_+ \geq n_- \geq 0. \tag{S1.10}$$

Each highest-weight class represents an $SU(3) \times SU(2)$ multiplet, with the product of the two irreducible dimensions.

Encode a letter with r, s derivatives by $u^r v^s$, so $J_+ = u\partial_v$. Its type and total order $r + s$ form a skeleton entry.

Group cochains by multisets of these entries. Compute each exact dimension by gauge-group Weyl integration and highest-weight extraction for $SU(3) \times SU(2)$.

For $SU(2)$ and $SU(3)$, combine flavour, dotted and colour tableaux.

The flavour slots are $\gamma^I \mapsto e_I$, $\beta_I \mapsto e_{I+1} \wedge e_{I+2}$, $\mathsf{b} \mapsto e_1 \wedge e_2 \wedge e_3$, and the empty slot for c.

The dotted tableau uses n_- brackets $[pq] = u_p v_q - v_p u_q$ and $n_+ - n_-$ further u factors. Contract the colour slots into trace words.

Match the spanning-set rank to the exact dimension. For $SU(3)$, split by charge conjugation $C : t_A \mapsto -t_A^{\mathsf{T}}$.

For $SU(N)$ with $N \geq 4$, use $V = \mathbb{C}^{3|2}$ and the $\mathfrak{gl}(3|2)$ Young diagrams μ with $\mu_4 \leq 2$. Its coordinates are $\theta_1, \theta_2, \theta_3$ and the two holomorphic coordinates.

A diagram with $|\mu| = \Theta + n_+ + n_-$ gives

$$\lambda(\mu) = (\mu_1, \mu_2, \mu_3;\ \#\{i \geq 4 : \mu_i \geq 1\},\ \#\{i \geq 4 : \mu_i \geq 2\}), \qquad 2(E + J_L) = 2(\mu_1 + \mu_2 + \mu_3) + 3\sum_{i \geq 4} \mu_i, \tag{S1.11}$$

S1.3 Evaluation at finite rank

We evaluate trace words or invariant tensors on Lie-algebra-valued letters of the chosen gauge group, so that finite-rank identities are imposed directly. The same evaluations turn the graph rule for each loop correction into a matrix acting between the specified charge sectors.

Evaluate trace words, or invariant tensors for SO, Sp and G_2, on random Lie-algebra-valued letters modulo $p = 2^{31} - 1$.

These evaluations impose the finite-rank trace identities. Rank equal to the exact sector dimension certifies the spanning set.

For $SU(N)$, use traceless $N \times N$ matrices and $\kappa_{RS} = \mathrm{tr}(t_R t_S)$, $t^R = \kappa^{RS} t_S$.

$$\sum_R (t_R)_{ab} (t^R)_{cd} = \delta_{ad}\delta_{bc} - \frac{1}{N}\delta_{ab}\delta_{cd}.$$

For $SO(2n+1)$ or $Sp(n)$, impose $A^{\mathsf{T}}F + FA = 0$, with $\bar{\imath} = N_F + 1 - i$. Use $F_{i\bar{\imath}} = 1$, $N_F = 2n+1$, or $F_{i\bar{\imath}} = s_i$, $N_F = 2n$, where $s_i = 1$ for $i \leq n$ and -1 otherwise.

$$\sum_R (t_R)_{ab} (t^R)_{cd} = \tfrac{1}{2}\delta_{ad}\delta_{bc} - \tfrac{1}{2}F_{ac}F_{bd}. \tag{S1.12}$$

Apply (7.3) by selecting $n + 1$ input slots and replacing them with two differentiated output letters, using the selected Taylor coefficient.

The colour chain becomes nested commutators.

$$Q_n F \ \ni\ F(\ldots, \underbrace{[P, t_{R_1}]}_{\text{slot } i_1}, \ldots, \underbrace{[t^{R_1}, t_{R_2}]}_{\text{slot } i_2}, \ldots, \underbrace{[t^{R_n}, Q]}_{\text{slot } i_{n+1}}, \ldots), \tag{S1.13}$$

Sum over ordered slot choices. Here P, Q are the output letters, and the Casimir tensor contracts each R_j.

S1.4 Ranks of the successive differentials

We compute successive obstructions by solving the correction equations and measuring the residuals modulo all allowed lower-order corrections. Choosing bases for boundaries, cohomology representatives, and their primitives makes this calculation explicit and yields the rank tests below.

For $d_r^k : E_r^k \to E_r^{k+1}$, choose boundary columns B_r^k, extend them to cycle columns $(B_r^k\ H_r^k)$, and choose primitive columns L_r^k satisfying

$$d_r^k L_r^k = B_r^{k+1}, \qquad (B_r^k\ H_r^k\ L_r^k) \text{ is invertible.} \tag{S1.14}$$

The corresponding contraction is

$$\begin{aligned} i_r^k &= H_r^k, \\ \pi_r^k &= (0\ 1\ 0)(B_r^k\ H_r^k\ L_r^k)^{-1}, \\ h_r^k &= L_r^{k-1}(1\ 0\ 0)(B_r^k\ H_r^k\ L_r^k)^{-1}. \end{aligned} \tag{S1.15}$$

The block 1 is the identity on the indicated columns. These matrices satisfy

$$\begin{aligned} &d_r^{k-1}h_r^k + h_r^{k+1}d_r^k = 1 - i_r^k\pi_r^k, \qquad \pi_r^k i_r^k = 1, \\ &h_r^{k-1}h_r^k = 0, \qquad h_r^k i_r^k = 0, \qquad \pi_r^{k-1}h_r^k = 0. \end{aligned} \tag{S1.16}$$

Pivot splitting. Order the monomials and reduce the columns of d_r^k over $\mathbb{Q}$. New-pivot preimages form L_r^k, their images form B_r^{k+1}, and independent kernel columns modulo B_r^k form H_r^k.

For large sectors, select pivots after modular row projection, reconstruct their coefficients over $\mathbb{Q}$, and verify them by exact Q_0 application.

Orthogonal splitting. With orthonormal monomials, take $H_r^k = \ker d_r^k \cap (\operatorname{im} d_r^{k-1})^\perp$, orthogonal projection π_r, and the Moore–Penrose pseudoinverse h_r of d_r.

Decoupled pair. For $Q_0\beta^s = -m\gamma_s'$ and $Q_0\gamma_s' = 0$ in (S8.2), replace each differentiated γ_s' by the same derivative of $-\beta^s/m$, with the odd-derivation sign, and divide by the number of γ_s', β^s letters. The projection sets these letters to zero.

Two contractions give conjugate transferred differentials by (8.25).

$$\hbar\,\pi_r'(\hbar) i_r(\hbar)\,\mathbf{Q}^{(r+1)} = \pi_r'(\hbar)\,\mathbf{Q}^{(r)}\,i_r(\hbar) = \hbar\,\mathbf{Q}'^{(r+1)}\,\pi_r'(\hbar) i_r(\hbar), \tag{S1.17}$$

The constant term $\pi_r' i_r$ is invertible on E_{r+1}, so the ranks and class vanishing are independent of the splitting.

Check nilpotence and the inclusion-intertwining identity through the retained order in $\hbar$.

Let $\mathcal{K}$ be a basis of source Q_0-cycles in $(\lambda)_Y$.

Corrections lie in $C_m = C^{Y-m}(\lambda - m\mathsf{d})$ and order-k residuals in $C_k' = C^{Y+1-k}(\lambda - k\mathsf{d})$.

Assemble the source and correction columns into

$$\begin{aligned} M_R = \Big[\ \underbrace{Q_k\mathcal{K}}_{\text{source cocycles}}\ \Big|\ \underbrace{Q_{k-m} : C_m \to C_k'}_{\text{corrections}}\ \Big]_{1\le m\le k\le R}, \qquad M_R' = M_R \setminus \{Q_k\mathcal{K}\}, \\ \operatorname{rank} d_R = (\operatorname{rank} M_R - \operatorname{rank} M_{R-1}) - (\operatorname{rank} M_R' - \operatorname{rank} M_{R-1}'), \end{aligned} \tag{S1.18}$$

Block row k lies in C'_k, with $M_0 = M'_0 = 0$.

Proposition S1.3.

Equation (S1.18) computes $\operatorname{rank} d_R$ on $E_R^Y(\lambda)$.

Proof

$$\operatorname{rank}\begin{pmatrix} A & 0 \\ B & C \end{pmatrix} = \operatorname{rank} A + \operatorname{rank}\left[B|_{\ker A} \;\; C\right].$$

For M'_2, row elimination leaves $\operatorname{im} Q_0 + Q_1 \ker Q_0$ in C'_2, with cycles taken in C_1. For M_2, it also leaves $Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1$ for every pair satisfying $Q_1\mathcal{O}_0 + Q_0\mathcal{O}_1 = 0$. Their rank difference is the quotient test (S1.19).

Exact sources give residuals in this denominator. For $Q_0 z = Q_0 y = 0$,

$$\begin{aligned} \mathcal{O}_0 = Q_0 x, \quad \mathcal{O}_1 = Q_1 x + z \quad &\Longrightarrow \quad Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1 = -Q_0 Q_2 x + Q_1 z, \\ \mathcal{O}_0 = Q_1 y, \quad \mathcal{O}_1 = Q_2 y + z \quad &\Longrightarrow \quad Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1 = -Q_0 Q_3 y + Q_1 z. \end{aligned}$$

The same row elimination through R reproduces (8.39). □

The second-page test is

$$d_2[\mathcal{O}_0]_2 = 0 \quad \Longleftrightarrow \quad Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1 \in \operatorname{im} Q_0 + Q_1 \ker Q_0, \tag{S1.19}$$

Take $\ker Q_0$ in the first-correction sector, so it includes every cycle added to $\mathcal{O}_1$.

A rational functional annihilating this denominator and nonzero on the residual certifies $d_2[\mathcal{O}_0]_2 \neq 0$, as in S8.1.

The surviving dimension is

$$\dim E_{r+1}^k = \dim E_r^k - \operatorname{rank} d_r^k - \operatorname{rank} d_r^{k-1}. \tag{S1.20}$$

For candidate cycles $v_1, \ldots, v_m \in \ker d_r^k$, their independent classes number

$$\operatorname{rank}(B_r^k \; v_1 \; \cdots \; v_m) - \operatorname{rank} B_r^k = \operatorname{rank}\left[\pi_r^k(v_1 \; \cdots \; v_m)\right]. \tag{S1.21}$$

Reconstruct representatives with $i_0(\hbar) \cdots i_r(\hbar)$.

Evaluate block rows at probe points and reduce modulo p, using the pivot splitting.

For a single class, solve each correction and check the residual directly.

$$R_r = \sum_{k=1}^{r} Q_k \mathcal{O}_{r-k}, \qquad Q_0\mathcal{O}_r = -R_r, \qquad R_r \in \operatorname{im} Q_0.$$

S1.5 Cartan restriction and fortuity

We perform two separate tests: Cartan restriction detects non-Cartan classes, while comparison with stable cohomology detects fortuitous classes. Both tests reduce to ranks of linear maps, with finite-rank trace identities supplying an alternative computation of the fortuitous count.

Cartan restriction sets the Q_0 commutators to zero, so it descends to E_1 with rank ρ.

Its kernel consists of non-Cartan classes [3, Definition 1].

Fortuitous classes are independent modulo the image of stable cohomology at rank N [3, 4].

Let C_∞ be the stable trace complex, $\pi_N : C_\infty \to C_N$ its rank restriction, and $R_N = \ker \pi_N$ its trace identities.

With $B_N = \operatorname{im} Q_0$ and $h = \dim E_1$, compute

$$f = h - \operatorname{rank} \pi_N^*, \qquad \operatorname{rank} \pi_N^* = \operatorname{rank} \begin{pmatrix} \pi_N & B_N \\ Q_0^\infty & 0 \end{pmatrix} - \operatorname{rank} Q_0^\infty - \operatorname{rank} B_N, \tag{S1.22}$$

The sequence $0 \to R_N \to C_\infty \to C_N \to 0$ identifies degree-$Y-1$ fortuity with $K(Y) = \dim \ker[H^Y(R_N) \to H^Y(C_\infty)]$. In the formula below, Q is the restriction of Q_0^∞ to R_N, and ι is its inclusion.

$$\begin{aligned} f(Y-1) = K(Y) = \dim R^Y - \operatorname{rank} \begin{pmatrix} Q : R^Y \to R^{Y+1} & 0 \\ \iota : R^Y \to C_\infty^Y & Q_0 : C_\infty^{Y-1} \to C_\infty^Y \end{pmatrix} \\ + \operatorname{rank} Q_0|_{C_\infty^{Y-1}} - \operatorname{rank} Q|_{R^{Y-1}}. \end{aligned} \tag{S1.23}$$

The block kernel consists of $Qz = 0$, $\iota z = Q_0 w$, with $z \in R^Y$ and $w \in C_\infty^{Y-1}$. Subtract the freedom $\ker Q_0$ in w and the boundaries in R.

For $SU(N \geq 4)$, compute R_N diagram by diagram using (S1.11).

$SU(N)$ trace identities start at $N+1$ letters. The sector with Y letters is stable for $N \geq Y+1$.

The $SO(2n+1)$ and $Sp(n)$ stable trace algebras agree by trace reversal. Count them by the Burnside action of $\prod_k S_2 \wr S_k$ on perfect matchings. The stable ranges are respectively $Y < 2n+2$ and $Y < n+1$, where $f = 0$.

S1.6 Selection rules and the all-order criterion

At fixed charges, the grading permits only finitely many incoming and outgoing differentials. We combine these bounds with the graviton completion theorem to determine which obstruction tests are needed for all-order survival.

Theorem 8.1 makes every outgoing d_r vanish on the graviton part. The source ranks obey the fortuitous-rank bound (8.51).

For all-order survival, check every outgoing map with $r \leq \min_I \theta_I$ vanishes on the class and every incoming map with $r \leq \min(n_+, n_-)$ misses it.

These finite tests give a closed completion by (8.39).

The scans compute (h, ρ) sectorwise and d_1, d_2, d_3 on pairs with nonzero source and target pages. For $SU(N)$, they compute (h, f) and follow the fortuitous classes found.

S1.7 Certification and gates

We record which rank computations are certified over the rational numbers and which are supported by independent finite-field probes. Nilpotence, deliberate perturbations of the graph rules, and low-rank group identifications provide additional checks on the calculation.

Evaluate ranks modulo $p = 2^{31} - 1$.

The modular rank bounds the rational rank from below. Equality with the source or target dimension certifies the rank.

The label [p,p′] records independent probes at $p' = 2147483629$. The label [$\mathbb{Q}$] records rational elimination or Chinese-remainder reconstruction beyond the Hadamard bound for the minors.

Check $\sum_{j=0}^n Q_j Q_{n-j} = 0$ for $n \leq 3$. Perturbing a loop coefficient, vertex sign or projection must fail these checks.

Compare the adjoint counts of $SO(3), Sp(1), SU(2)$ and of $SO(5), Sp(2)$, then compare orbit sums with full torus sums sectorwise.

S1.8 A worked chain

We apply the preceding tests to the $SU(2)$ class P39, following its obstruction chain through every order allowed by the grading. The table records the target sectors and residual calculations used to establish its all-order survival.

Worked chain P39

$$(\lambda)_Y = (5,5,5;3,0)_9, \qquad E = 15, \qquad R_I = 2, \qquad J_L = \frac{9}{2}, \qquad J_R = \frac{3}{2}.$$

For the $SU(2)$ class P39, $n_- = 0$ excludes incoming maps and Corollary S1.2 bounds outgoing maps by $r \leq 5$.

Solve the residuals in the following target sectors.

r	target $(\lambda - r\mathsf{d})_{Y+1-r}$	target data	result
1	$(4,4,4;4,1)_9$	$h = 13$	$d_1 = 0$
2	$(3,3,3;5,2)_8$	583 graviton classes	$R_2 \in \operatorname{im} Q_0$ (45,450,714 terms)
3	$(2,2,2;6,3)_7$	$h = 1$	$R_3 \in \operatorname{im} Q_0$ (17,787,235 terms)
4	$(1,1,1;7,4)_6$	$h = \rho = g = 152$	$R_4 \in \operatorname{im} Q_0$ (1,276,152 terms)
5	$(0,0,0;8,5)_5$	$h = 0$	no target

Here g counts graviton classes, and the term counts refer to the residuals.

The completion is closed through four loops. Its fifth target is zero, giving all-order survival [p,p′].

S2 N=4 Yang–Mills theory with gauge group SU(N)

We apply the census and lifting tests to the special unitary gauge groups. The tables record the completed search ranges, the fortuitous classes found, and their successive obstructions or closed quantum completions.

S2.1 Frontier

We first specify which charge sectors have been searched at each rank. The census separates completed ranges from partial scans and records the cohomology and fortuitous-class counts within them.

Census over $\mathbb{F}_p$.

G	all sectors through $E + J_L$	partial levels (sectors done)	fortuitous classes found
$SU(2)$	$\frac{33}{2}$	$\frac{35}{2}$ by the restriction detector; chains at $\frac{39}{2}$, 20	$\mathcal{O}_0$ and its descendants, $\mathcal{O}_1$, P39, X40
$SU(3)$	15	detector through 17	X_7, the towers F_1, B_1, four further primaries
$SU(4)$	$\frac{27}{2}$	14: 34 of 38 diagrams	one class at $(4,2,2;2,2)_9$
$SU(5)$	14	$\frac{29}{2}$: 26/29; 15: 3/53	none
$SU(6)$	15	$\frac{31}{2}$: 1/38	none
$SU(7)$	15	$\frac{31}{2}$: 19/37; 16: 14/62	none
$SU(8)$	$\frac{31}{2}$	16: 45/56; $\frac{33}{2}$: 19/45	none

G	all sectors through $E + J_L$	partial levels (sectors done)	fortuitous classes found
$SU(9)$	$\frac{33}{2}$	17: 41/64; $\frac{35}{2}$: 18/50	none
$SU(10)$	17	$\frac{35}{2}$: 33/38; 18: 39/72	none
$SU(11)$–$SU(15)$	18		none

The earlier $SU(3)$ scan reached $E + J_L = \frac{23}{2}$ [5].

$E + J_L$	sectors	classes	boson states	fermion states	non-graviton blocks	non-graviton classes
1	1	0	0	0	0	0
$\frac{3}{2}$	1	0	0	0	0	0
2	2	1	6	0	0	0
$\frac{5}{2}$	1	1	0	6	0	0
3	5	2	1	8	0	0
$\frac{7}{2}$	2	2	18	0	0	0
4	6	4	18	12	0	0
$\frac{9}{2}$	5	4	2	38	0	0
5	9	6	36	30	0	0
$\frac{11}{2}$	6	7	102	18	0	0
6	16	15	59	139	0	0
$\frac{13}{2}$	9	17	96	228	0	0
7	19	29	423	114	0	0
$\frac{15}{2}$	16	37	402	420	0	0
8	26	60	345	912	0	0
$\frac{17}{2}$	19	72	1230	714	0	0
9	39	112	1786	1173	0	0
$\frac{19}{2}$	26	140	1542	2934	0	0
10	46	211	3327	3507	0	0
$\frac{21}{2}$	39	271	6618	3734	0	0
11	59	413	6807	8733	0	0
$\frac{23}{2}$	46	531	9582	13824	0	0
12	81	811	21483	13593	1	1
$\frac{25}{2}$	59	1053	26406	25614	1	1
13	94	1563	30504	46380	3	3
$\frac{27}{2}$	81	2028	63444	49640	4	5
14	116	2986	90111	74934	7	9
$\frac{29}{2}$	94	3835	101250	138786	7	11
15	150	5567	177657	170608	12	19
$\frac{31}{2}$	116	7203	280314	222792	9	19
16	172	10330	333117	391821	14	27
$\frac{33}{2}$	150	13377	494548	548270	15	30

$SU(2)$ counts are highest weights, with $SU(3) \times SU(2)$ multiplicities for states. Seven-letter classes through level $\frac{33}{2}$ belong to $\mathcal{O}_0$. Nine-letter classes at 16, $\frac{33}{2}$ are its graviton products and descendants.

S2.2 Fortuitous classes

We list the fortuitous classes by their charges and highest-weight sectors. The rank calculation for the first $SU(4)$ example shows how the restriction detector isolates a class beyond the stable spectrum.

G	class	$(\lambda)_Y$	E	(R_1, R_2, R_3)	J_L	J_R	data
$SU(2)$	$\mathcal{O}_0$	$(4,4,4;0,0)_7$	$\frac{19}{2}$	$(\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{5}{2}$	0	[6, 7, Section 3.1]
	$\mathcal{O}_1$	$(6,6,6;0,0)_9$	$\frac{27}{2}$	$(\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{9}{2}$	0	[4]
	P39	$(5,5,5;3,0)_9$	15	$(2,2,2)$	$\frac{9}{2}$	$\frac{3}{2}$	$h=13$: 12 products and descendants, one new class [p,p′]
	X40	$(7,5,5;1,1)_{10}$	$\frac{31}{2}$	$(\frac{7}{2},\frac{3}{2},\frac{3}{2})$	$\frac{9}{2}$	0	$(h,\rho,g)=(8,8,7)$ [p,p′]
$SU(3)$	X_7	$(2,2,2;2,2)_7$	$\frac{21}{2}$	$(\frac{5}{2},\frac{5}{2},\frac{5}{2})$	$\frac{3}{2}$	0	$(h,\rho,g)=(26,26,25)$; odd under C; 35 highest-weight blocks in its multiplet
	X_7 member	$(0,0,0;5,5)_7$	$\frac{27}{2}$	$(\frac{7}{2},\frac{7}{2},\frac{7}{2})$	$\frac{3}{2}$	0	fortuitous [$\mathbb{Q}$]; 26 trace generators
	F_1 tower	$(n+4,4,4;0,0)_{n+7}$	$n+\frac{19}{2}$	$(n+\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{5}{2}$	0	$n\geq 3$ [5, 8]
	B_1 tower	$(n+5,5,4;0,0)_{n+8}$	$n+11$	$(n+2,2,1)$	3	0	[5, 8]
	primary	$(5,4,4;1,1)_{10}$	$\frac{27}{2}$	$(\frac{7}{2},\frac{5}{2},\frac{5}{2})$	$\frac{5}{2}$	0	$h=2$, $\rho=0$, odd under C
	primary	$(7,4,4;1,0)_{11}$	14	$(5,2,2)$	$\frac{5}{2}$	$\frac{1}{2}$	$h=2$ (one beyond the F_1 descendant), $\rho=0$
	primary	$(6,4,4;1,1)_{11}$	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{5}{2}$	0	$h=3$, $\rho=0$
	primary	$(7,5,4;1,0)_{11}$	$\frac{29}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{3}{2})$	3	$\frac{1}{2}$	$h=4$, $\rho=0$
$SU(4)$	first class	$(4,2,2;2,2)_9$	$\frac{25}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{3}{2}$	0	$f=1$ [$\mathbb{Q}$]; odd under C

C denotes charge conjugation $t_A \mapsto -t_A^{\mathsf{T}}$. $(0,0,0;5,5)_7$ represents a nonzero class of $H^7(\mathfrak{sl}_3[z_+,z_-],\mathfrak{sl}_3)$ over $\mathbb{Q}$. Raising operators give fixed-Y shifts $(-e_I;+e_\pm)$ and $(+e_I;0)$, e.g. $(7,4,4;1,0)_{11} \to (6,4,4;1,1)_{11}$.

Apply (S1.23) to the $SU(4)$ diagram $(4,2,2,2,2)$.

Y	5	6	7	8	9	10	11	12
$\dim R^Y$	17	213	1201	3430	5321	4579	2085	404
$\operatorname{rank} Q\|_{R^Y}$	17	196	1005	2424	2894	1681	401	0
$\dim H^Y(R_4)$	0	0	0	1	3	4	3	3
$K(Y)$	0	0	0	0	0	1	0	0

At $Y=10$, $K = 4579 - 7900 + 6216 - 2894 = 1$ over $\mathbb{Q}$, hence $f(9)=1$. The same diagram for $SU(5)$ and the other completed $SU(4)$ diagrams have $K=0$.

S2.3 Lifting

We follow the listed classes through the successive obstruction equations. The tables give the target sectors, exactness tests and incoming differentials used to determine whether each class survives.

class	result	level
$\mathcal{O}_0$ $(SU(2))$	closed completion $\sum_{r\le 4}\hbar^r\mathcal{O}_r$, no incoming d_r: all orders (Section S8.3)	$[\mathbb{Q}]$
$\mathcal{O}_1$ $(SU(2))$	$d_1 \neq 0$ [7, Section 3.3]	[p,p′]
P39	$d_1 = 0$; $R_2, R_3, R_4 \in \operatorname{im} Q_0$; $r = 5$ target $h = 0$; no incoming: all orders	[p,p′]
X40	$d_1 = 0$; $R_2 \in \operatorname{im} Q_0 + Q_1 \ker Q_0$; $R_3, R_4 \in \operatorname{im} Q_0$; $[R_5] = 0$; incoming sources $h = 0$: all orders	[p,p′] for $r \le 2$, [p]
X_7	$d_1 = 0$ in and out; outgoing d_2 target and incoming d_2 source have $h = 0$; $r \ge 3$ empty: all orders	[p]
F_1, $n = 3$	closed completion $\mathcal{O}_0, \dots, \mathcal{O}_4$; $r \ge 5$ empty; no incoming: all orders	[p]
$(5,4,4;1,1)_{10}$	$d_1 = d_2 = 0$ on both classes; the d_3 target has $h = 0$; no incoming	[p]
$SU(4)$, $(4,2,2;2,2)_9$	R_1 exact in $(3,1,1;3,3)_9$ $(h = 111)$; R_2 exact $(h = 0)$; $r \ge 3$ empty; incoming d_1 rank 0 from 453 sources, incoming d_2 source $h = 0$: all orders	[p]

class	r	target	data	result
X40	1	$(6,4,4;2,2)_{10}$		$d_1 = 0$ (4,429,603 terms)
	2	$(5,3,3;3,3)_9$		$R_2 \in \operatorname{im} Q_0 + Q_1 \ker Q_0$
	3	$(4,2,2;4,4)_8$	1112 graviton classes	$R_3 \in \operatorname{im} Q_0$
	4	$(3,1,1;5,5)_7$		$R_4 \in \operatorname{im} Q_0$
	5	$(2,0,0;6,6)_6$	$h = \rho = g = 30$	restriction of R_5 zero
F_1, $n = 3$, $(7,4,4;0,0)_{10}$	1	$(6,3,3;1,1)_{10}$	$h = 0$	$\mathcal{O}_1$ in $Y = 9$
	2	$(5,2,2;2,2)_9$	$h = 609$, $\rho = 608$	$\mathcal{O}_2$ in $Y = 8$
	3	$(4,1,1;3,3)_8$	$h = 0$	$\mathcal{O}_3$ in $Y = 7$
	4	$(3,0,0;4,4)_7$	$h = \rho = 34$	$\mathcal{O}_4$ in $Y = 6$
$SU(4)$, $(4,2,2;2,2)_9$	1	$(3,1,1;3,3)_9$	$h = 111$	R_1 exact
	2	$(2,0,0;4,4)_8$	$h = 0$	R_2 exact
	in 1	$(5,3,3;1,1)_9$	$h = 453$	incoming d_1 rank 0
	in 2	$(6,4,4;0,0)_{10}$	$h = 0$	no incoming d_2
$(5,4,4;1,1)_{10}$	1	$(4,3,3;2,2)_{10}$	$h = 0$	R_1 exact on both classes
	2	$(3,2,2;3,3)_9$	$h = 331$, $\rho = 329$	R_2 exact on both classes

The 17 second-page classes in seven eleven-letter $SU(2)$ blocks are products.

S3 SO(2N+1), Sp(N) and S-duality

We compare the orthogonal and symplectic censuses to locate the classical classes responsible for unequal counts. We then apply the lifting tests and compare the surviving spectra at fixed physical charges.

S-duality predicts equality at each charge for $SO(2N+1)$ and $Sp(N)$. The quantities Y, n_+ and n_- are not conserved charges. The first mismatch, at $SO(7), Sp(3)$ [9, 10], is lifted by d_1 [11].

S3.1 First fortuitous levels

We locate the first fortuitous levels in each orthogonal and symplectic theory. The tables give the charges and lifting status of these classes alongside the census counts from which they were found.

G	$(\lambda)_Y$	E	(R_1, R_2, R_3)	J_L	data	lifted or not
$SO(3) = Sp(1)$	$(4,4,4;0,0)_7$	$\frac{19}{2}$	$(\frac{3}{2},\frac{3}{2},\frac{3}{2})$	$\frac{5}{2}$	the $SU(2)$ class $\mathcal{O}_0$	all orders survival
$SO(5) = Sp(2)$	$(3,2,2;1,1)_7$	9	$(3,2,2)$	1	$f=1$, $\rho=0$	all orders [p]
$SO(7)$	$(3,3,3;0,0)_8$	$\frac{17}{2}$	$(\frac{5}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	O_f [9]; $(h,\rho)=(1,1)$	d_1 into $(2,2,2;1,1)_8$ [11]
$Sp(3)$	$(5,2,2;1,1)_9$	11	$(5,2,2)$	1	$h=f=1$	all orders [p]
$Sp(3)$	$(5,0,0;3,3)_9$	13	$(7,2,2)$	1	$h=1$, $\rho=0$	no outgoing; incoming d_2 open
$SO(9)$	$(5,3,3;0,0)_{10}$	$\frac{21}{2}$	$(\frac{9}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	$h=6$, one beyond $Sp(4)$	d_1 into $(4,2,2;1,1)_{10}$ [p]
$SO(11)$	$(7,3,3;0,0)_{12}$	$\frac{25}{2}$	$(\frac{13}{2},\frac{5}{2},\frac{5}{2})$	$\frac{1}{2}$	$h=23$, $f=1$	d_1 into $(6,2,2;1,1)_{12}$ [p]

For $N=3,4,5$, the first $SO(2N+1)$ class has $E=\frac{4N+5}{2}$, $J_L=\frac{1}{2}$, and the lift $(2N-3,3,3;0,0)_{2N+2} \xrightarrow{d_1} (2N-4,2,2;1,1)_{2N+2}$. For $Sp(3)$, $(5,2,2;1,1)_9 \to (5,0,0;3,3)_9$ has shift $(0,-2,-2;+2,+2)$ at fixed Y. For $Sp(3)$ sector $(5,2,2;1,1)_9$ and $SO(5) = Sp(2)$, outgoing targets have $E_r = 0$ or zero rank, and incoming sources have $h=0$. Entries give $\sum h$/fortuitous count. An asterisk excludes stable-range sectors from h, and superscript p marks a partial level.

$E+J_L$	$SO(3)$	$Sp(1)$	$SO(5)$	$Sp(2)$	$SO(7)$	$Sp(3)$	$SO(9)$	$Sp(4)$	$SO(11)$	$Sp(5)$
8	60	60	164	164	43*	203	0*	207	0*	207
$\frac{17}{2}$	72	72	207	207	117*	255	0*	261	0*	261
9	112	112	381	381	385/1*	495	0*	516	0*	516
$\frac{19}{2}$	140	140	495	495	596/1*	647	0*	675	0*	675
10	211	211	877/1	877/1	1218/3*	1230	168*	1304	0*	1309
$\frac{21}{2}$	271	271	1151/2	1151/2	1641/5*	1636	554*	1736	0*	1745
11	413	413	1996/6	1996/6	3047/9	3029	2058/1*	3277	0*	3308
$\frac{23}{2}$	531	531	2665/9	2665/9	4099/11	4077	3627/2*	4426	0*	4470
12	811/1	811/1	90/1p	90/1p	100p	7408/1	0p	8252	651*	8374
$\frac{25}{2}$	–	–	–	–	–	1385p	–	1748p	1642p	1821p
13	261p	261p	726/7p	726/7p	868p	1380p	842p	1659p	848/1p	1741p
$\frac{27}{2}$	–	–	–	–	–	24p	–	44p	50p	50p

$E+J_L$	$SO(3)$	$Sp(1)$	$SO(5)$	$Sp(2)$	$SO(7)$	$Sp(3)$	$SO(9)$	$Sp(4)$	$SO(11)$	$Sp(5)$
14	160^{p}	160^{p}	$378/6^{\mathrm{p}}$	$378/6^{\mathrm{p}}$	429^{p}	424^{p}	434^{p}	442^{p}	13^{p}	447^{p}

S3.2 One-loop lifts

We calculate the one-loop differentials connecting the extra classical classes to their targets. The source and target ranks show which orthogonal classes are removed and how the corresponding symplectic sectors differ.

source	target	result	level
$SO(7)$ $(3,3,3;0,0)_8$, O_f	$(2,2,2;1,1)_8$, $(h,\rho)=(2,1)$	d_1 rank 1 [11]	[p], three primes
$SO(7)$ $(4,3,3;0,0)_8$	$(3,2,2;1,1)_8$, $(h,\rho)=(11,10)$	d_1 rank 1	[p]
$SO(7)$ $(3,1,1;2,2)_8$	$(2,0,0;3,3)_8$, $(h,\rho)=(2,1)$	$d_1\neq 0$; for $Sp(3)$ $d_1=0$	[$\mathbb{Q}$]
$SO(9)$ $(5,3,3;0,0)_{10}$	$(4,2,2;1,1)_{10}$, $(h,\rho)=(10,9)$	d_1 rank 1	[p]
$SO(11)$ $(7,3,3;0,0)_{12}$	$(6,2,2;1,1)_{12}$	d_1 rank 1	[p]
$SO(7)$ $(6,1,1;2,2)_{10}$, $h-\rho=1$, $f=0$	from $(7,2,2;1,1)_{10}$	incoming d_1 rank 0; incoming d_2 source $h=0$	[p]

sector	(h,ρ) for $SO(2N+1)$	(h,ρ) for $Sp(N)$
$(3,3,3;0,0)_8$	$(1,1)$	$(0,0)$
$(2,2,2;1,1)_8$	$(2,1)$	$(1,1)$
$(4,3,3;0,0)_8$	$(7,7)$	$(6,6)$
$(3,2,2;1,1)_8$	$(11,10)$	$(10,10)$
$(5,3,3;0,0)_{10}$	$(6,6)$	$(5,5)$
$(4,2,2;1,1)_{10}$	$(10,9)$	$(9,9)$

Along the d-orbit $(5,0,0;3,3)_9$, $(6,1,1;2,2)_{10}$, $(7,2,2;1,1)_{11}$, $(8,3,3;0,0)_{12}$, the h/ρ values are $0/0, 355/354, 1/0, 41/40$ for $SO(7)$ and $1/0, 354/354, 1/0, 41/40$ for $Sp(3)$.

S3.3 The per-charge S-duality test

We test S-duality by collecting the surviving classes at each fixed set of physical charges. The comparison sums over all sectors with those charges and records the remaining candidates for higher differentials.

Assuming $E_2=E_\infty$ and perturbative S-duality, at every charge $(E;R_I;J_L,J_R)$,

$$\sum_{(\lambda)_Y\mapsto(E;R_I;J_L,J_R)}\dim E_2^{SO(2N+1)}(\lambda)_Y=\sum_{(\lambda)_Y\mapsto(E;R_I;J_L,J_R)}\dim E_2^{Sp(N)}(\lambda)_Y, \tag{S3.1}$$

The sums run over fixed-charge sectors related by $k\mathsf{d}$ in Proposition S1.1. A mismatch requires a higher d_r or a nonperturbative lift.

$E+J_L$	$\sum h^{SO}$	$\sum h^{Sp}$	charges with $h^{SO}\neq h^{Sp}$	sectors with $d_1\neq 0$ (SO/Sp)	$\sum \dim E_2$	charges with $E_2^{SO}\neq E_2^{Sp}$	d_3 pairs
$SO(7)$ and $Sp(3)$							
$\leq \frac{17}{2}$	equal		0	0/0	equal	0	—
9	497	495	2	1/0	495	0	—
$\frac{19}{2}$	649	647	2	1/0	647	0	—
10	1236	1230	6	3/0	1230	0	2, rank 0
$\frac{21}{2}$	1646	1636	7	4/0	1636	0	2, rank 0
11	3047	3029	12	7/0	3029	0	5, rank 0
$SO(9)$ and $Sp(4)$							
$\leq \frac{19}{2}$	equal		0	0/0	equal	0	—
10	1304	1304	0	0/0	1304	0	2, rank 0
$\frac{21}{2}$	1736	1736	0	0/0	1736	0	2, rank 0
11	3279	3277	2	1/0	3277	0	5, rank 0

Through level 11, E_2 agrees charge by charge. The d_2 candidate set is empty, and all d_3 ranks vanish [3].

S4 Commuting schemes and non-Cartan classes

We construct invariant functions on commuting tuples that vanish on a Cartan subalgebra but remain nonzero in the commuting-scheme quotient. Their Gram determinants account for the additional orthogonal classes and give explicit counterexamples to the cited Cartan-restriction claims.

S4.1 Setting

We set up the quotient by the commutator relations and its restriction to a Cartan subalgebra. The Gram determinant and the chiral-ring theorem provide the two ingredients for detecting invariant classes that this restriction misses.

Work over algebraically closed k of characteristic zero, with connected reductive G, Lie algebra $\mathfrak{g}$, Cartan subalgebra $\mathfrak{h}$ and $W = N_G(\mathfrak{h})/Z_G(\mathfrak{h})$. For a tuple $(x_1,\dots,x_d)$ of $d\geq 1$ Lie-algebra elements, the commutator ideal $I_d = ([x_i,x_j])_{i<j}$ defines $k[\mathcal{C}_{\mathfrak{g}}^d] = k[\mathfrak{g}^d]/I_d$ and

$$\Phi:\ k[\mathcal{C}_{\mathfrak{g}}^d]^G \longrightarrow k[\mathfrak{h}^d]^W. \tag{S4.1}$$

Chen–Ngô Conjecture 3.1 asserts that Φ is an isomorphism, equivalently a reduced normal quotient.

Conjecture 3.2 implies injectivity [12, Section 3]. The isomorphism holds for GL_n [12–14, Theorem 3.3] and Sp_{2n} [15, Theorem 4.1], and is asserted for O_n, SO_n [16, Theorem 1.2]. It also holds for $d=2$ [17, Theorem 1.4.11, 47, 48]. Reduced-scheme restriction is an isomorphism in types A–D and G_2 [18, 19].

Fix a nondegenerate invariant symmetric form B.

$$T(x) = \big(B(x_i,x_j)\big)_{1\leq i,j\leq d}, \qquad \det T \in k[\mathfrak{g}^d]^G_{2d}. \tag{S4.2}$$

For $g\in GL_d$, linear recombination of $(x_1,\dots,x_d)^{\mathsf{T}}$ by g sends its Gram matrix to gTg^{T}.

$$\det(gTg^{\mathsf{T}}) = (\det g)(\det T)(\det g^{\mathsf{T}}) = (\det g)^2 \det T.$$

Weight $\det^2$ means that this recombination multiplies $\det T$ by $(\det g)^2$.

For $d > \operatorname{rk}\mathfrak{g}$, $\det T|_{\mathfrak{h}^d} = 0$. For three adjoint scalars and the trace form, $\det T = \frac{1}{6}\varepsilon_{IJK}\varepsilon_{LMN}\operatorname{tr}(\gamma^I\gamma^L)\operatorname{tr}(\gamma^J\gamma^M)\operatorname{tr}(\gamma^K\gamma^N)$.

Definition S4.1.

For finite-dimensional super E, set $\mathcal{O}(\mathcal{C}_E) = \operatorname{Sym}(E\otimes\mathfrak{g}^*)/J_E$, where J_E is generated by $\xi([\theta(e),\theta(e')])$, $e,e'\in E$, $\xi\in\mathfrak{g}^*$. Its R-points are even $\theta\in\operatorname{Hom}(E,\mathfrak{g})\otimes R$ satisfying these equations. Restriction gives $\Phi_E:\mathcal{O}(\mathcal{C}_E)^G\to\mathcal{O}(\operatorname{Hom}(E,\mathfrak{h}))^W$.

$E=k^d$ gives the commuting scheme, and $E=k^{d_B|d_F}$ gives the super-commuting scheme [3, Section 2]. Use the following external chiral-ring theorem.

Theorem S4.2 (Cachazo–Douglas–Seiberg–Witten)**.**

Let $\mathfrak{g}$ be simple with dual Coxeter number $h^\vee$. For a basis (e_a) and odd generators X^a, Y^a, put $X=\sum_a e_a\otimes X^a$, $Y=\sum_a e_a\otimes Y^a$ and $A=\Lambda(X^a,Y^a)/(\{X,X\},\{X,Y\},\{Y,Y\})$, with $\{X,Y\}=\sum_{a,b}[e_a,e_b]\otimes X^aY^b$. For $S=\sum_{a,b}B(e_a,e_b)X^aY^b$, (i) $A^{\mathfrak{g}}=k[S]$, (ii) $S^{h^\vee}=0$, and (iii) $S^{h^\vee-1}\neq 0$.

The theorem was conjectured in [20]. Proofs cover (i) in [21], classical types in [20, 22, 23], $\mathfrak{g}_2$ in [24], and the remaining exceptional types in [25]. $A=\mathcal{O}(\mathcal{C}_{\Pi k^2})$, with Πk^2 purely odd of dimension $0|2$.

S4.2 The transfer

We test whether the Gram determinant survives the commutator quotient by evaluating the commuting variables in the chiral-ring algebra. The resulting determinant identity reduces its nonvanishing to the known powers of the chiral-ring generator.

Lemma S4.3.

For the exterior algebra $\Lambda=\Lambda[u_1,\dots,u_d,v_1,\dots,v_d]$ on odd generators, set $N_{ij}=u_iv_j+u_jv_i$.

$$\det(N_{ij})_{1\le i,j\le d} = (d+1)!\,u_1v_1u_2v_2\cdots u_dv_d. \tag{S4.3}$$

Proof.

For odd $\theta_i,\bar\theta_i$, Berezin integration extracts $\bar\theta_1\theta_1\cdots\bar\theta_d\theta_d$.

$$\frac{1}{d!}\left(\sum_{i,j}\bar\theta_iM_{ij}\theta_j\right)^d = \det M\,\bar\theta_1\theta_1\cdots\bar\theta_d\theta_d,\qquad \det M=\int e^{\bar\theta M\theta}\quad (M\in\operatorname{Mat}_d(\Lambda_{\bar0})).$$

Put $\alpha=\sum_i\bar\theta_iu_i$, $\beta=\sum_j v_j\theta_j$, $\gamma=\sum_i\bar\theta_iv_i$, $\varepsilon=\sum_j u_j\theta_j$. These are even, and $\bar\theta N\theta=\alpha\beta-\gamma\varepsilon$, so $\det N=\int\sum_{r+s=d}\frac{(-1)^s}{r!s!}\alpha^r\gamma^s\beta^r\varepsilon^s$.

$$\alpha^r\gamma^s = r!s!\sum_{I\sqcup J=[d],\ |I|=r}\prod_{i\in I}\bar\theta_iu_i\prod_{j\in J}\bar\theta_jv_j,$$

$$\beta^r\varepsilon^s = r!s!\sum_{K\sqcup L=[d],\ |K|=r}\prod_{i\in K}v_i\theta_i\prod_{j\in L}u_j\theta_j,$$

$$\text{nonzero terms}\iff K=I,\ L=J.$$

$$\bar\theta_iu_iv_i\theta_i=\bar\theta_i\theta_iu_iv_i,\qquad \bar\theta_jv_ju_j\theta_j=-\bar\theta_j\theta_ju_jv_j.$$

$$\det N=\sum_{r=0}^d\frac{(-1)^s}{r!s!}(r!s!)^2\binom{d}{r}(-1)^s\prod_m u_mv_m=\sum_{r=0}^d d!\prod_m u_mv_m=(d+1)!\prod_m u_mv_m,\qquad s=d-r.$$

□

Proposition S4.4.

For simple $\mathfrak{g}$, set $R_d = (A \otimes \Lambda)_{\bar{0}}$ and $x_i = Xu_i + Yv_i$. Then $[x_i, x_j] = 0$, $B(x_i, x_j) = -S(u_i v_j + u_j v_i)$, and evaluation gives $\varphi(\det T) = (-1)^d (d+1)! S^d \otimes u_1 v_1 \cdots u_d v_d$.

Proof.

Moving the odd coefficients past X, Y gives $[x_i, x_j] = -\sum_{a,b} [e_a, e_b] \otimes (X^a X^b u_i u_j + X^a Y^b u_i v_j + Y^a X^b v_i u_j + Y^a Y^b v_i v_j)$.

$$\{X, X\} = \{X, Y\} = \{Y, X\} = \{Y, Y\} = 0 \quad \Longrightarrow \quad [x_i, x_j] = 0.$$

With $B_{ab} = B(e_a, e_b)$, the XX and YY terms vanish by symmetry, while $\sum B_{ab} X^a Y^b = S$, $\sum B_{ab} Y^a X^b = -S$ and $v_i u_j = -u_j v_i$. Hence $B(x_i, x_j) = -S(u_i v_j + u_j v_i)$.

$$\varphi(I_d) = 0, \qquad \varphi(\det T) = \det(-SN_{ij}) = (-S)^d \det N.$$

□

Corollary S4.5.

If $d < h^\vee$, then $\det T \notin I_d$.

Proof.

For $d < h^\vee$, Theorem S4.2 gives $S^d \neq 0$. Freeness over A gives $S^d \otimes u_1 v_1 \cdots u_d v_d \neq 0$, hence $\varphi(\det T) \neq 0$.

$$\varphi(I_d) = 0, \qquad \varphi(\det T) \neq 0 \quad \Longrightarrow \quad \det T \notin I_d.$$

□

S4.3 The Gram component

We separate dependence on the space of field labels from dependence on the Lie algebra. This identifies the full representation containing the Gram determinant and determines exactly when its multiplicity is one or zero.

Lemma S4.6.

For $n \geq 0$, a $G \times S_n$-module V_n independent of E satisfies $\mathcal{O}(\mathcal{C}_E)_n \cong (E^{\otimes n} \otimes V_n)_{S_n}$. The S_n action on $E^{\otimes n}$ uses Koszul signs.

Proof.

$\mathrm{Sym}^n(E \otimes \mathfrak{g}^*) = (E^{\otimes n} \otimes (\mathfrak{g}^*)^{\otimes n})_{S_n}$. For the transposed bracket $c : \mathfrak{g}^* \to (\mathfrak{g}^*)^{\otimes 2}$, set $K_2 = c(\mathfrak{g}^*)$. The relations are the classes of $e \otimes e' \otimes c(\xi)$.

$$K_n = \langle S_n \cdot (K_2 \otimes (\mathfrak{g}^*)^{\otimes (n-2)}) \rangle, \qquad (J_E)_n = (E^{\otimes n} \otimes K_n)_{S_n}.$$

Exactness of coinvariants in characteristic zero gives $V_n = (\mathfrak{g}^*)^{\otimes n} / K_n$ and the asserted quotient. □

For a Specht module Sp_λ, $\lambda \vdash n$, super Schur–Weyl duality gives $\mathbb{S}_\lambda(E) = (E^{\otimes n} \otimes \mathrm{Sp}_\lambda)_{S_n}$ [26, 27]. At $E = k^{m|p}$, it is nonzero exactly when $\lambda_{m+1} \leq p$. These modules are irreducible and pairwise non-isomorphic, with $\mathbb{S}_\lambda(\Pi k^e) \cong \mathbb{S}_{\lambda'}(k^e)$. Here λ' is the transpose partition.

$$V_n = \bigoplus_{\lambda \vdash n} \mathrm{Sp}_\lambda \otimes M_\lambda, \qquad M_\lambda = \mathrm{Hom}_{S_n}(\mathrm{Sp}_\lambda, V_n).$$

$$\mathcal{O}(\mathcal{C}_E)_n = \bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(E) \otimes M_\lambda, \qquad \mathcal{O}(\mathcal{C}_E)_n^G = \bigoplus_{\lambda \vdash n} \mathbb{S}_\lambda(E) \otimes M_\lambda^G. \tag{S4.4}$$

Lemma S4.7.

There are maps $\phi_\lambda : M_\lambda^G \to \mathbb{S}_\lambda(\mathfrak{h}^*)^W$, independent of E, such that $\Phi_E = \bigoplus_\lambda \mathrm{id}_{\mathbb{S}_\lambda(E)} \otimes \phi_\lambda$. Components with more than $\operatorname{rk} \mathfrak{g}$ parts restrict to zero.

Proof.

Restriction induces $(\mathfrak{g}^*)^{\otimes n} \to (\mathfrak{h}^*)^{\otimes n}$ and kills K_n because $\mathfrak{h}$ is abelian. Apply $(E^{\otimes n} \otimes -)_{S_n}$ and use $\operatorname{Hom}_{S_n}(\mathrm{Sp}_\lambda, (\mathfrak{h}^*)^{\otimes n}) = \mathbb{S}_\lambda(\mathfrak{h}^*)$. Taking G-invariants gives ϕ_λ. If λ has more than $\dim \mathfrak{h}$ parts, $\mathbb{S}_\lambda(\mathfrak{h}^*) = 0$. □

Theorem S4.8 (Gram component)**.**

Let $\mathfrak{g}$ be simple, G connected and $d \geq 1$. For every finite-dimensional super E, the multiplicity space L_d is independent of E and

$$\mathcal{O}(\mathcal{C}_E)_{(2^d)}^G \cong \mathbb{S}_{(2^d)}(E) \otimes L_d, \qquad \dim L_d = \begin{cases} 1, & d < h^\vee, \\ 0, & d \geq h^\vee, \end{cases} \tag{S4.5}$$

Here (2^d) has d parts. For $E = k^d$, the degree-$2d$ SL_d-invariants are spanned by $\det T$ when $d < h^\vee$.

Proof.

Equation (S4.4) gives $L_d = M_{(2^d)}^G$. At $E = \Pi k^2$, degree-$2d$ SL_2-invariance selects $\lambda' = (d, d)$, giving $(A_{2d}^G)^{SL_2} \cong L_d$. Theorem S4.2 gives $(A_{2d}^G)^{SL_2} = kS^d$, since $S \mapsto \det(g)S$. Its dimension is 1 for $d < h^\vee$ and 0 otherwise. At $E = k^d$, SL_d-invariance selects $\mathbb{S}_{(2^d)}(k^d) = \det^2$. The nonzero $\det T$ of Corollary S4.5 spans it. □

S4.4 Cartan restriction and nilpotency

We compare the Gram component with its restriction to a Cartan subalgebra. The calculation identifies the range in which the determinant is nonzero in the quotient but vanishes at every ordinary commuting tuple, hence is nilpotent.

Lemma S4.9.

For semisimple $\mathfrak{g}$ and commuting $x_i = s_i + n_i$, the semisimple parts s_i share a Cartan subalgebra, and $f(x) = f(s)$ for every invariant polynomial f.

Proof.

Since $\operatorname{ad} s_i, \operatorname{ad} n_i$ are polynomial in $\operatorname{ad} x_i$, all s_i, n_j commute. The s_i span a toral subalgebra. In $L = Z_G(s_1, \ldots, s_d)^\circ$, all n_i lie in a common Borel nilradical. A cocharacter positive on this nilradical gives $\mu(t) \cdot x_i \to s_i$ and $f(x) = f(\mu(t) \cdot x) \to f(s)$. See [19], [18, Section 6] and [12, Section 3]. □

Corollary S4.10.

For simple $\mathfrak{g}$ and $\operatorname{rk} \mathfrak{g} < d < h^\vee$, $\det T$ is nonzero nilpotent with $\Phi(\det T) = 0$. This contradicts Conjectures 3.1 and 3.2 of [12].

Proof.

Corollary S4.5 gives $\det T \notin I_d$.

$$d > \dim \mathfrak{h} \quad \Longrightarrow \quad \Phi(\det T) = 0.$$

Lemma S4.9 gives $\det T(x) = \det T(s) = 0$ at every k-point. The Nullstellensatz gives $\det T \in \sqrt{I_d}$, proving nilpotency. Both conjectures require injectivity [12, Section 3]. □

For $d \le \operatorname{rk}\mathfrak{g}$, $\Phi : L_d \to \mathbb{S}_{(2^d)}(\mathfrak{h}^*)^W$ is an isomorphism. The Gram determinant is nonzero, and Steinberg's irreducibility gives $\dim \operatorname{Sym}^2(\Lambda^d\mathfrak{h}^*)^W = 1$ [28, Chapter V, Section 2, Exercise 3]. The range $\operatorname{rk}\mathfrak{g} < d < h^\vee$ is

type	rk	$h^\vee$	d with $\operatorname{rk} < d < h^\vee$
A_n	n	$n+1$	—
B_n, $n \ge 2$	n	$2n-1$	$n+1 \le d \le 2n-2$ $(n \ge 3)$
C_n, $n \ge 3$	n	$n+1$	—
D_n, $n \ge 4$	n	$2n-2$	$n+1 \le d \le 2n-3$
G_2	2	4	$d = 3$
F_4	4	9	$5 \le d \le 8$
E_6	6	12	$7 \le d \le 11$
E_7	7	18	$8 \le d \le 17$
E_8	8	30	$9 \le d \le 29$

S4.5 Two examples

We work out the nonzero restriction-kernel classes for G_2 and $SO(7)$. Explicit trace identities and commutator ranks distinguish these classes from Gram determinants that vanish in the quotient.

For multilinear $m(a_1, a_2, a_3, b_1, b_2, b_3)$, define the antisymmetrization

$$\langle m \rangle = \sum_{\sigma,\tau \in S_3} \operatorname{sgn}\sigma \, \operatorname{sgn}\tau \, m\big(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\tau(1)}, x_{\tau(2)}, x_{\tau(3)}\big), \tag{S4.6}$$

For the trace form, $\langle \operatorname{tr}(a_1b_1)\operatorname{tr}(a_2b_2)\operatorname{tr}(a_3b_3)\rangle = 6 \det T$.

Example S4.11 ($\mathfrak{g}_2$, $d = 3$)**.**

Here $2 < 3 < 4 = h^\vee$. Set $Z_3 = (\mathbb{S}_{(2,2,2)}\mathfrak{g}_2)^{G_2}$, $Y_3 = (\mathfrak{g}_2 \otimes \mathbb{S}_{(2,1,1)}\mathfrak{g}_2)^{G_2}$. Reductivity and Pieri's rule give $I_3 \cap Z_3 = \operatorname{im} Y_3$. Exact dimensions are $4, 3$. Lemma S4.7 and Theorem S4.8 give $\Phi(Z_3) = 0$, $Z_3/(I_3 \cap Z_3) = k \det T$.

Use traces in the seven-dimensional representation.

$$\begin{aligned} S_{6a} &= \langle \operatorname{tr}(a_1a_2a_3b_1b_2b_3)\rangle, & S_{6b} &= \langle \operatorname{tr}(a_1b_1a_2b_2a_3b_3)\rangle, \\ F_{42} &= \langle \operatorname{tr}(a_1a_2b_1b_2)\operatorname{tr}(a_3b_3)\rangle, & F_{33} &= \langle \operatorname{tr}(a_1a_2a_3)\operatorname{tr}(b_1b_2b_3)\rangle. \end{aligned} \tag{S4.7}$$

$$Z_3 = \operatorname{span}\{S_{6a}, S_{6b}, F_{42}, \det T\}, \qquad I_3 \cap Z_3 = \operatorname{span}\{S_{6a}, S_{6b}, F_{42}\},$$
$$-18S_{6a} - 6S_{6b} + 3F_{42} + 4F_{33} = 0.$$

Through degree 10, $\ker\Phi = k \det T$ in type $(2,2,2)$. Moreover $(\det T)^2 \in I_3$, since the 30-dimensional $(\mathbb{S}_{(4,4,4)}\mathfrak{g}_2)^{G_2}$ lies in the commutator image. For $\mathfrak{sl}_3, \mathfrak{so}_5, \mathfrak{sp}_4$, rk = 2 and

$h^\vee = 3$. With defining traces, the explicit certificates for $\det T \in I_3$ are

$$\begin{aligned}
\mathfrak{sl}_3: \ \det T = \langle &-\tfrac{5}{6}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{2}{3}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
&+ \tfrac{5}{18}\operatorname{tr}([a_1,a_2]a_3)\operatorname{tr}(b_1b_2b_3)\rangle, \\
\mathfrak{sp}_4: \ \det T = \langle &-\tfrac{1}{6}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{5}{6}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
&+ \tfrac{1}{4}\operatorname{tr}([a_1,a_2]b_1b_3)\operatorname{tr}(a_3b_2)\rangle, \\
\mathfrak{so}_5: \ \det T = \langle &\tfrac{10}{3}\operatorname{tr}([a_1,a_2]a_3b_1b_2b_3) - \tfrac{4}{3}\operatorname{tr}([a_1,a_2]b_1a_3b_2b_3) \\
&- \tfrac{2}{3}\operatorname{tr}([a_1,a_2]a_3)\operatorname{tr}(b_1b_2b_3) \\
&+ \operatorname{tr}([a_1,a_2]b_1b_3)\operatorname{tr}(a_3b_2)\rangle.
\end{aligned} \tag{S4.8}$$

Each term contains a commutator and lies in I_3.

Example S4.12 ($\mathfrak{so}_7$, $d = 4$)**.**

Here $3 < 4 < 5 = h^\vee$. The O_7-invariant $\det T$ is nonzero nilpotent with zero restriction. The target and commutator-source dimensions are $9, 12$. Rank is at least 8 modulo three primes and at most 8 by Theorem S4.8, hence exactly 8.

Set $Z_d = (\mathbb{S}_{(2^d)}\mathfrak{g})^G$, $Y_d = (\mathfrak{g} \otimes \mathbb{S}_{(2^{d-2},1,1)}\mathfrak{g})^G$ for the target and source of commutator substitution.

$\mathfrak{g}$, d	$(\dim Y_d, \dim Z_d)$	rank	$\det T \in I_d$	$\dim L_d$	$\Phi(\det T)$
$\mathfrak{sl}_3$, 3	$(4,3)$	3	yes	0	—
$\mathfrak{so}_5, \mathfrak{sp}_4$, 3	$(4,4)$	4	yes	0	—
$\mathfrak{g}_2$, 3	$(3,4)$	3	no	1	0
$\mathfrak{so}_7, \mathfrak{sp}_6$, 3	$(4,5)$	4	no	1	$\neq 0$
$\mathfrak{sl}_4$, 3	$(5,5)$	≥ 4	no	1	$\neq 0$
$\mathfrak{so}_5$, 4	$(9,5)$	5	yes	0	—
$\mathfrak{g}_2$, 4	$(7,5)$	5	yes	0	—
$\mathfrak{sl}_4$, 4	$(20,8)$	8	yes	0	—
$\mathfrak{sp}_6$, 4	$(12,8)$	8	yes	0	—
$\mathfrak{so}_7$, 4	$(12,9)$	≥ 8	no	1	0

Corollary S4.5 proves the nonzero classes in lower-bound rows. Other ranks are exact, including all $d = 3$ ranks over $\mathbb{Z}$ and a second octonionic $\mathfrak{g}_2$ realization.

S4.6 The orthogonal case of Song, Xia and Xu

We check the Pfaffian lemma used in the orthogonal Cartan-restriction argument of Song, Xia and Xu. An explicit rank bound contradicts its required nonvanishing and identifies the orthogonal conclusions affected by that failure.

The orthogonal proof of [16, Theorem 1.2] uses Theorem 3.4, Proposition 3.1 and Lemmas 3.2–3.3. Lemma 3.3 depends on Lemmas 5.3 and 5.2. Even SO_{2m} uses the O_{2m} theorem. The symplectic proof uses the Chen–Ngô Pfaffian independently. For generic skew (t_{ij}) of size $2d+3$, write $h_{i_1\cdots i_k}$ for the Pfaffian with those indices deleted, and zero for repeated indices. The $2d \times 2d$ matrix is

$$\mathsf{A}_{ij} = h^2_{i,j,2d+2} + h^2_{i,2d+1,2d+2} + h^2_{j,2d+1,2d+2}, \qquad 1 \leq i,j \leq 2d. \tag{S4.9}$$

[16, Lemma 5.2] asserts a common solution of $h_{2d+2} = 0$ and $h_{2d+3}\det \mathsf{A} \neq 0$.

Theorem S4.13.

On $h_{2d+2} = 0$, $\operatorname{rk} \mathsf{A} \le 3$. Thus $\det \mathsf{A} \in (h_{2d+2})$ for $d \ge 2$, contradicting [16, Lemma 5.2].

Proof.

Deleting index $2d+2$ gives P of size $2d+2$, with $\operatorname{Pf} P = h_{2d+2}$ and final original index $2d+3$. Define $P^{\sharp}_{ab} = (-1)^{a+b} \operatorname{Pf} P(\hat{a}, \hat{b})$ for $a < b$ and $P^{\sharp}_{ba} = -P^{\sharp}_{ab}$. Pfaffian expansion gives $PP^{\sharp} = (\operatorname{Pf} P)\mathbf{1}$, $h_{i,j,2d+2} = \pm P^{\sharp}_{ij}$ and $h_{i,2d+1,2d+2} = \pm P^{\sharp}_{i,2d+1}$. On $\operatorname{Pf} P = 0$, $\operatorname{im} P^{\sharp} \subseteq \ker P$. Rank $2d$ gives $\dim \ker P = 2$. Rank at most $2d-2$ gives $P^{\sharp} = 0$. Thus $P^{\sharp} = vw^{\mathsf{T}} - wv^{\mathsf{T}}$. Squaring gives $h^2_{i,j,2d+2} = (v_i w_j - v_j w_i)^2$ and $h^2_{i,2d+1,2d+2} = \ell_i^2$, with $\ell_i = w_{2d+1} v_i - v_{2d+1} w_i$. Put $f_i = (v_i^2, w_i^2, v_i w_i, 1)$, $a = w^2_{2d+1}$, $b = v^2_{2d+1}$, $c = -2 w_{2d+1} v_{2d+1}$. Then $\mathsf{A}_{ij} = f_i M f_j^{\mathsf{T}}$, where

$$M = \begin{pmatrix} 0 & 1 & 0 & a \\ 1 & 0 & 0 & b \\ 0 & 0 & -2 & c \\ a & b & c & 0 \end{pmatrix}, \tag{S4.10}$$

$$\det M = -2 \det \begin{pmatrix} 0 & 1 & a \\ 1 & 0 & b \\ a & b & 0 \end{pmatrix} + c^2 = -4ab + c^2 = 0.$$

$$\operatorname{rk} \mathsf{A} \le \operatorname{rk} M \le 3 < 2d.$$

The Pfaffian h_{2d+2} is irreducible [29, Lemma B.2.10]. The Nullstellensatz gives $\det \mathsf{A} \in (h_{2d+2})$. □

The identities also hold in characteristic $p > 2$, invalidating [16, Lemma 5.2] and its orthogonal consequences for $d \ge 2$. The induction in [16, Lemma 5.2] asserts degree $4d-4$ in t_{12} with a prescribed product coefficient. The leading rows of $\mathsf{A}_{1j}, \mathsf{A}_{2j}$ coincide, cancelling t_{12}^{4d}. At $d = 2, 3$, the exact degree is $4d-2$, and integer specializations contradict the proposed t_{12}^{4d-4} coefficient. At $d = 1$, $\det \mathsf{A} = -(t_{35}^2 + (t_{25} + t_{15})^2)(t_{35}^2 + (t_{25} - t_{15})^2)$, consistent with the base case and $2d \le 3$. On sampled rational points of $h_{2d+2} = 0$, the ranks are 2 for $d = 1$ and 3 for $d = 2, 3, 4$.

Corollary S4.14.

For $n = 2m+1 \ge 7$, $m+1 \le d \le 2m-2$, or $n = 2m \ge 8$, $m+1 \le d \le 2m-3$, both O_n, SO_n have nonzero $\ker \Phi$ and invariant nilradical. This contradicts [16, Theorem 1.2(1), Corollary 1.3] at $n = 7, d = 4$ and Lemmas 3.3,5.3 in the stated orthogonal ranges.

Proof.

Apply Corollary S4.10 in types B_m, D_m. The certificate $\det T$ is also O_n-invariant. The independent Proposition 3.1 and Lemma 3.2 of [16] force Lemma 3.3 to fail. □

S4.7 Super-commuting schemes and N=4 theory

We translate the commuting-scheme results into the classical cohomology of the Yang–Mills theory. The resulting highest-weight sectors identify the exceptional and orthogonal non-Cartan classes, while the symplectic restriction remains injective.

Chang–Yin [30] identify classical cohomology with $H^{\bullet}(\mathfrak{g}[\mathsf{A}], \mathfrak{g}; \mathbb{C})$, where $\mathsf{A} = \mathbb{C}[z_+, z_-] \otimes \Lambda(\theta_1, \theta_2, \theta_3)$. At top degree $p = |n|$, $n = (n_{z_+}, n_{z_-}, n_{\theta_1}, n_{\theta_2}, n_{\theta_3})$, this is $\mathcal{O}(\mathcal{C}_{k^{3|2}})^G_n$ [3, Proposition 4]. The θ_i-components are even and $z_\pm$-components odd. This is sector $(n_{\theta_1}, n_{\theta_2}, n_{\theta_3}; n_{z_+}, n_{z_-})$ with $Y = p$ in Appendix S1.1.

Corollary S4.15.

For simple $\mathfrak{g}$ and $1 \le d < h^{\vee}$, the component $\mathbb{S}_{(2^d)}(k^{3|2}) \otimes L_d$ is nonzero of degree $2d$. For

the even-first Borel, its highest weight is $((2^{\min(d,3)}, 0^{3-\min(d,3)}) \mid (c,c))$, $c = \max(d-3, 0)$. The highest-weight sectors are $(2,0,0;0,0)_2$, $(2,2,0;0,0)_4$, and $(2,2,2;d-3,d-3)_{2d}$ for $d = 1, 2, \geq 3$. Its $(0,0,0;d,d)_{2d}$ part is kS^d, with $S = B(X,Y)$. For $d > \operatorname{rk}\mathfrak{g}$, the component lies in $\ker \Phi_{k^{3|2}}$.

Proof.

Apply the hook condition $\lambda_4 \leq 2$ and highest-weight formula $(\lambda_1, \lambda_2, \lambda_3 \mid \max(\lambda_1' - 3, 0), \max(\lambda_2' - 3, 0))$ to $\lambda = (2^d)$ [27]. Even-coordinate degree zero gives $\mathcal{O}(\mathcal{C}_{\Pi k^2}) = A$ and $\mathbb{S}_{(2^d)}(\Pi k^2) = \det^d$. Theorem S4.8 gives $(A^G_{2d})^{SL_2} = kS^d$, and Lemma S4.7 gives the restriction statement. □

Corollary S4.16.

If every Φ_{k^d} is an isomorphism, so is Φ_E for every finite-dimensional super vector space E. For $\mathfrak{g} = \mathfrak{sp}_{2N}$, this makes $\Phi_{k^{3|2}}$ an isomorphism.

Proof.

Every partition occurs in some k^d, so every ϕ_λ is invertible in $\Phi_{k^d} = \bigoplus_\lambda \mathrm{id} \otimes \phi_\lambda$. The same decomposition gives Φ_E. The symplectic hypothesis is [15, Theorem 4.1]. □

For $\mathfrak{g}_2$, $d = 3$ gives the non-Cartan class $\det T$ in $(2,2,2;0,0)_6$, protected to all orders in Appendix S5. For $\mathfrak{so}_7$, $d = 4$ gives $(2,2,2;1,1)_8$ at level 9, of dimension 85 versus 84 for $\mathfrak{sp}_6$ [10, Section 3, Table 1]. Corollary S4.15 spans its one-dimensional restriction kernel, the O_f target S3 [3, Theorem 1]. For $(\mathfrak{so}_{2N+1}, \mathfrak{sp}_{2N})$, $h^\vee = (2N-1, N+1)$. The orthogonal non-Cartan sectors are $(2,2,2;d-3,d-3)_{2d}$, $N+1 \leq d \leq 2N-2$, whereas the symplectic restriction is injective. At $N = 4$, these are $(2,2,2;2,2)_{10}$ and $(2,2,2;3,3)_{12}$, with $E = 12, 15$, $R_I = E/3$ and $J_L = J_R = 0$. The odd coordinates are $\lambda_\pm$ [31, Section 2]. For $\operatorname{rk}\mathfrak{g} < d < h^\vee$, $[S^d]$ is the non-Coulomb monotone class in the same $\mathfrak{gl}(3|2)$ module as $(2,2,2;d-3,d-3)_{2d}$ [31, Section 5].

S5 G_2

We apply the census to G_2 using its seven-dimensional representation. The calculation isolates a scalar class invisible on the Cartan and tests its perturbative survival alongside the remaining charge sectors.

S5.1 Realization

We first construct the matrices needed to evaluate G_2 cochains and loop corrections. Invariant dimensions and the first nilpotence identities check the representation and the resulting operators.

Realize $\mathfrak{g}_2 \subset \mathfrak{gl}(7)$ by preserving $\eta_{i\bar{\imath}} = 1$, $\bar{\imath} = 8 - i$, and the three-form

$$\begin{aligned} \phi &= e^{123} + 2\,e^{567} + e^{147} + e^{246} + e^{345}, \\ \sum_d &(A_{da}\phi_{dbc} + A_{db}\phi_{adc} + A_{dc}\phi_{abd}) = 0, \end{aligned} \tag{S5.1}$$

The Cartan elements are $\operatorname{diag}(a_1, a_2, -a_1 - a_2, 0, a_1 + a_2, -a_2, -a_1)$. Short roots are $\pm(1,0), \pm(0,1), \pm(1,1)$ and long roots are $\pm(1,-1), \pm(2,1), \pm(1,2)$.

$$\begin{gathered} \dim \operatorname{Sym}^k(\mathfrak{g}_2)^{G_2} = 1,1,2,2,2,3,3 \quad (k = 2,4,\ldots,14), \\ \dim \Lambda^k(\mathfrak{g}_2)^{G_2} = 1 \quad (k = 0,3,11,14), \qquad \dim \operatorname{Hom}_{G_2}(\mathfrak{g}_2^{\otimes 4}, \mathbb{C}) = 5. \end{gathered}$$

Check $Q_0Q_1 + Q_1Q_0 = 0$ in $(3,3,3;0,0)_5$, $(4,3,3;0,0)_6$, $(3,3,2;1,0)_5$, with 275, 5254, 7646 entries. Check $Q_0Q_2 + Q_1^2 + Q_2Q_0 = 0$ in $(4,4,4;0,0)_6$, $(4,4,3;1,0)_6$, with 32,186, 2,077,705 entries. Seven-dimensional trace words span the exact dimensions in every computed sector.

S5.2 The scalar class

We identify the extra scalar class by comparing invariant polynomials with classical boundaries. Its vanishing Cartan restriction and the available differential targets then determine its non-Cartan character and all-order survival.

In $(2,2,2;0,0)_6$, $E=6$, $R_I=2$, $J_L=J_R=0$. Use six-scalar singlets and $\beta\gamma^4$ boundaries. Cauchy and Pieri give $C^6=Z_3$, $C^5=Y_3$ of Example S4.11.

$\mathfrak{g}$	$\dim C^5$	$\dim C^6$	h	ρ	$\det T \in \operatorname{im} Q_0$
$\mathfrak{su}_3$	4	3	0	0	yes
$\mathfrak{so}_5=\mathfrak{sp}_4$	4	4	0	0	yes
$\mathfrak{g}_2$	3	4	1	0	no
$\mathfrak{so}_7$, $\mathfrak{sp}_6$	4	5	1	1	no
$\mathfrak{su}_4$	5	5	1	1	no

For $\mathfrak{g}_2$, the class is $[\det_{IJ}\operatorname{tr}_7(\gamma^I\gamma^J)]=[\frac{1}{6}\varepsilon_{IJK}\varepsilon_{LMN}\operatorname{tr}_7(\gamma^I\gamma^L)\operatorname{tr}_7(\gamma^J\gamma^M)\operatorname{tr}_7(\gamma^K\gamma^N)]$, with cochain and boundary ranks $4,3$ over $\mathbb{Q}$ and $\rho=0$. Trace types are $(2,2,2),(2,4),(3,3),(6)$. Commutator insertion gives the boundary space, with complement the $(2,2,2)$ class. $n_\pm=0$ excludes incoming differentials. Outgoing $d_1=0$ and the d_2 targets have $h=0$, giving all-order survival. Purely scalar blocks have $h=\rho$ for $SO(7),Sp(3)$ through level 11 and $SO(9),Sp(4)$ through $\frac{19}{2}$. For G_2 through 10, the exception is $(2,2,2;0,0)_6$. Its products with quadratic and quartic flavour covariants are exact.

S5.3 Census

The table collects the complete G_2 census through $E+J_L=\frac{23}{2}$. It records classical and non-Cartan dimensions together with the tested loop differentials.

Complete G_2 census through $E+J_L=\frac{23}{2}$.

$E+J_L$	sectors	$\sum h$	$\sum(h-\rho)$	$d_1\neq 0$	d_2 pairs	d_3 pairs
4	6	5	0	0	—	—
$\frac{9}{2}$	5	5	0	0	—	—
5	9	10	0	0	—	—
$\frac{11}{2}$	6	12	0	0	—	—
6	16	27	1	0	—	—
$\frac{13}{2}$	9	31	1	0	—	—
7	19	58	3	0	—	—
$\frac{15}{2}$	16	73	5	0	—	—
8	26	137	9	0	—	—
$\frac{17}{2}$	19	171	11	0	—	—
9	39	307	20	0	—	—
$\frac{19}{2}$	26	397	23	0	—	—
10	46	693	43	0	—	rank 0
$\frac{21}{2}$	39	911	62	0	—	rank 0
11	59	1564	120	0	—	rank 0
$\frac{23}{2}$	46	2075	170	0	—	rank 0

S6 Two-loop lifts in the Leigh–Strassler deformation

We use the deformed cubic vertex to find classes lifted at two loops in the Leigh–Strassler theory. The census is followed by tests of translation descendants, scheme dependence, parameter dependence and fortuity.

S6.1 The theory and its sectors

We specify the interaction and the charge sectors preserved by the Leigh–Strassler deformation. These data determine the fan operators and the symmetry multiplicities used in the census.

For the Leigh–Strassler deformation of $SU(3)$ [32], the cubic interaction is

$$\begin{aligned} &\frac{1+q}{4}\cdot\frac{i}{3!}(\varepsilon_{IJK}C_{ABC}-iS_{IJK}d_{ABC})(\gamma^I)^A(\gamma^J)^B(\gamma^K)^C,\\ &S_{IJK}=\rho\,|\varepsilon_{IJK}|+h_e\,\delta_{IJK},\qquad \rho=\frac{1-q}{1+q},\qquad h_e=\frac{2h}{1+q}, \end{aligned} \tag{S6.1}$$

$d_{ABC}=2\,\mathrm{tr}(t_A\{t_B,t_C\})$, $\mathrm{tr}(t_At_B)=\frac{1}{2}\delta_{AB}$, $\mathrm{tr}(t_At_Bt_C)=\frac{1}{4}(d_{ABC}+iC_{ABC})$. Set $\delta_{IJK}=1$ for $I=J=K$, zero otherwise. $(q,h)=(1,0)$ gives $\mathcal{N}=4$ and $h=0$ gives the β-deformation. For $SU(2)$, $d_{ABC}=0$. In each fan (7.3), replace the cubic scalar vertex by $\varepsilon_{IJK}[\,\cdot\,,\cdot\,]+S_{IJK}\{\,\cdot\,,\cdot\,\}_{\mathfrak{su}(3)}$, with $\{L,R\}_{\mathfrak{su}(3)}=\{L,R\}-\frac{2}{3}\,\mathrm{tr}(LR)\mathbf{1}$. The h_e term preserves $U(1)\times\mathbb{Z}_3$, giving sectors

$$\begin{aligned} (\Theta,c;n_+,n_-)_Y&=\bigoplus_{\theta_1+\theta_2+\theta_3=\Theta,\ \theta_2+2\theta_3\equiv c\ (3)} C^Y(\theta;n_+,n_-),\\ d_r&:(\Theta,c;n_\pm)_Y\to(\Theta-3r,c;n_\pm+r)_{Y+1-r}. \end{aligned} \tag{S6.2}$$

The conserved charges are E,J_L,J_R, $\sum_I R_I=\frac{1}{2}(3Y-\Theta)$ and c. Flavour cycling sends $c\mapsto c+\Theta$. For $\Theta\not\equiv 0 \pmod 3$, each rank unit at $c=0$ counts three states. For $\Theta\equiv 0$, compute $c=0,1$ and recover $c=2$ with $(\rho,h_e)\mapsto(-\rho,-h_e)$. Use $(\rho,h_e)=(\frac{12345}{678},\frac{4321}{97})$. Through level $\frac{17}{2}$, d_2 agrees at $(q,h)=(-\frac{3889}{4341},\frac{976546}{421077})$, $(4,\frac{85}{6})$, $(1,\frac{4321}{97})$, a second seed and p'. Check nilpotence through cubic order. Wrong vertex signs, $\mathfrak{su}(3)$ projection or h-weight violate it.

S6.2 The census

We enumerate the classical classes and successive differential ranks through $E+J_L=11$. The detailed source–target tables separate the direct two-loop action from the obstruction after the first correction has been included.

Complete census through $E+J_L=11$, including all candidate d_1,d_2,d_3 pairs.

$E+J_L$	sectors	$\sum h_{E_1}$	d_1 pairs	$d_1\neq 0$	d_2 pairs	$d_2\neq 0$	d_3 pairs	$d_3\neq 0$	$\sum h_{E_\infty}$
1	1	0	0	0	0	0	0	0	0
$\frac{3}{2}$	2	0	0	0	0	0	0	0	0
2	1	2	0	0	0	0	0	0	2
$\frac{5}{2}$	1	1	0	0	0	0	0	0	1
3	6	3	0	0	0	0	0	0	3
$\frac{7}{2}$	1	3	0	0	0	0	0	0	3
4	3	6	0	0	0	0	0	0	6
$\frac{9}{2}$	6	11	1	0	0	0	0	0	11
5	3	9	0	0	0	0	0	0	9

$E+J_L$	sectors	$\sum h_{E_1}$	d_1 pairs	$d_1\neq 0$	d_2 pairs	$d_2\neq 0$	d_3 pairs	$d_3\neq 0$	$\sum h_{E_\infty}$
$\frac{11}{2}$	3	14	1	0	0	0	0	0	14
6	12	56	3	1	0	0	0	0	54
$\frac{13}{2}$	3	26	2	0	0	0	0	0	26
7	6	54	3	1	1	1	0	0	46
$\frac{15}{2}$	12	136	6	3	1	0	0	0	118
8	6	88	6	1	1	1	0	0	84
$\frac{17}{2}$	6	143	5	2	2	1	0	0	121
9	20	460	14	9	3	0	0	0	382
$\frac{19}{2}$	6	267	8	4	2	2	0	0	239
10	10	460	12	6	4	1	0	0	396
$\frac{21}{2}$	20	1224	21	15	5	2	0	0	1012
11	10	962	18	11	5	4	0	0	752

The charge columns give $(E, J_L, J_R, \sum_I R_I)$.

$E+J_L$	source	target	source charges	target charges	rank d_2	origin
7	$(7,0;0,0)_5$	$(1,0;2,2)_4$	$(6,1,0,4)$	$(\frac{13}{2},\frac{1}{2},0,\frac{11}{2})$	1	P_{14}
8	$(8,0;0,0)_6$	$(2,0;2,2)_5$	$(7,1,0,5)$	$(\frac{15}{2},\frac{1}{2},0,\frac{13}{2})$	1	P_{16}
$\frac{17}{2}$	$(7,0;1,0)_5$	$(1,0;3,2)_4$	$(7,\frac{3}{2},\frac{1}{2},4)$	$(\frac{15}{2},1,\frac{1}{2},\frac{11}{2})$	1	$\partial_{++}P_{14}$
$\frac{19}{2}$	$(8,0;1,0)_6$	$(2,0;3,2)_5$	$(8,\frac{3}{2},\frac{1}{2},5)$	$(\frac{17}{2},1,\frac{1}{2},\frac{13}{2})$	1	$\partial_{++}P_{16}$
$\frac{19}{2}$	$(8,0;1,0)_7$	$(2,0;3,2)_6$	$(\frac{17}{2},1,\frac{1}{2},\frac{13}{2})$	$(9,\frac{1}{2},\frac{1}{2},8)$	1	P_{19}
10	$(7,0;2,0)_5$	$(1,0;4,2)_4$	$(8,2,1,4)$	$(\frac{17}{2},\frac{3}{2},1,\frac{11}{2})$	1	$\partial^2_{++}P_{14}$
$\frac{21}{2}$	$(9,c;1,0)_7$, $c=0,1$	$(3,c;3,2)_6$	$(9,\frac{3}{2},\frac{1}{2},6)$	$(\frac{19}{2},1,\frac{1}{2},\frac{15}{2})$	6 each	P_{21}
11	$(8,0;1,1)_7$	$(2,0;3,3)_6$	$(\frac{19}{2},\frac{3}{2},0,\frac{13}{2})$	$(10,1,0,8)$	5	
11	$(8,0;2,0)_6$	$(2,0;4,2)_5$	$(9,2,1,5)$	$(\frac{19}{2},\frac{3}{2},1,\frac{13}{2})$	1	$\partial^2_{++}P_{16}$
11	$(8,0;2,0)_7$	$(2,0;4,2)_6$	$(\frac{19}{2},\frac{3}{2},1,\frac{13}{2})$	$(10,1,1,8)$	5	contains $\partial_{++}P_{19}$
11	$(8,0;2,0)_8$	$(2,0;4,2)_7$	$(10,1,1,8)$	$(\frac{21}{2},\frac{1}{2},1,\frac{19}{2})$	1	

P_m labels source level $m = 2(E+J_L)$. At $(7,0;0,0)_5$, $\dim E_1 = 5$, $\operatorname{rank} d_1 = 3$, $\dim E_2 = 2$, $\operatorname{rank} d_2 = 1$, $\dim E_\infty = 1$. Since $Q_2\mathcal{O}_0 = 0$, $d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2$ (8.33).

$E+J_L$	source	target	$\dim E_2$ source	$\dim E_2$ target	$Q_2\mathcal{K}\neq 0$	rank d_2
7	$(7,0;0,0)_5$	$(1,0;2,2)_4$	2	2	0	1
$\frac{15}{2}$	$(6,0;1,0)_5$	$(0,0;3,2)_4$	2	1	0	0
8	$(8,0;0,0)_6$	$(2,0;2,2)_5$	3	3	0	1
$\frac{17}{2}$	$(7,0;1,0)_5$	$(1,0;3,2)_4$	2	4	2112	1
$\frac{17}{2}$	$(7,0;1,0)_6$	$(1,0;3,2)_5$	3	1	0	0
9	$(6,0;2,0)_5$	$(0,0;4,2)_4$	4	2	5680	0
9	$(9,0;0,0)_7$	$(3,0;2,2)_6$	8	4	0	0
9	$(9,1;0,0)_7$	$(3,1;2,2)_6$	6	3	0	0

The column $Q_2\mathcal{K}\neq 0$ counts nonzero entries of the direct action on source cocycles. At levels $7, 8$, the lift is $[Q_1\mathcal{O}_1]_2$.

S6.3 Translation towers

We compare lifted classes with their holomorphic translation descendants. The induced maps identify which census entries belong to the same tower and which require new source classes.

Proposition S6.1.

∂_{++} raises n_+, commutes with Q_n, d_r, and maps $E_r(\Theta, c; n_+, n_-)_Y \to E_r(\Theta, c; n_+ + 1, n_-)_Y$. The displayed maps are injective [p].

Proof.

Translation commutes with (7.3) and J_+, preserving highest weights. Homogeneous contractions transfer this commutation to d_r by (8.23). The matrix check passes for $n \leq 2$ and fails for an altered derivation coefficient.

map	E_1	E_2	E_∞
$(7,0;0,0)_5 \to (7,0;1,0)_5$	$5 \to 5$, isomorphism	$2 \to 2$, isomorphism	$1 \to 1$, isomorphism
$(7,0;1,0)_5 \to (7,0;2,0)_5$	$5 \to 5$, isomorphism	$2 \to 2$, isomorphism	$1 \to 1$, isomorphism
$(8,0;0,0)_6 \to (8,0;1,0)_6$	$4 \to 6$, injective	$3 \to 3$, isomorphism	$2 \to 2$, isomorphism
$(1,0;2,2)_4 \to (1,0;3,2)_4$	$2 \to 4$, injective	$2 \to 4$, injective	$1 \to 3$, injective
$(1,0;3,2)_4 \to (1,0;4,2)_4$	$4 \to 6$, injective	$4 \to 6$, injective	$3 \to 5$, injective
$(2,0;2,2)_5 \to (2,0;3,2)_5$	$3 \to 11$, injective	$3 \to 11$, injective	$2 \to 10$, injective

□

For $J_R = 0$, the SL_2 highest-weight translation descendants are ∂_{++}^k, giving $n_- = 0$ below level 11. P_{19} has predecessor $h = 0$. The rank-6 P_{21} has predecessors with $d_2 = 0$. P_{21} matches P_{14} times the six $\operatorname{tr}\Phi^{(I}\Phi^{J)}$ primaries, giving rank 6 per c.

S6.4 Independence of the two-loop scheme

We test whether the first two-loop lifts change under an allowed modification of the two-loop operator. Charge bounds and the classical kernel in the target show that the stated symmetry-preserving changes vanish on these sources.

$$Q_2 \longmapsto Q_2 + \delta, \qquad [Q_0, \delta] = 0, \qquad \delta \mod [Q_0, X].$$

The induced change of d_2 is $[\delta\mathcal{O}_0]_2$.

Proposition S6.2.

Let δ have the charges of Q_2, preserve gauge, $SU(2)_R$, clock and shift symmetries, and act on at most three letters. At levels $7, 8, \frac{17}{2}$, every source has $\delta\mathcal{O}_0 = 0$, fixing d_2 from Q_0, Q_1 and nilpotence.

Proof.

Proposition S1.1 gives $\Delta\Theta = -6$, so a contributing subword must have $\Theta \geq 6$. Source words are $\gamma^4\mathsf{b}, \gamma^3\beta^2$ and $\gamma^5\mathsf{b}, \gamma^4\beta^2$, whose three-letter subwords have $\Theta \leq 5$. Hence $\delta = 0$ on these source spaces, as is Q_2. At $(7,0;1,0)_5$, candidates are $\partial\mathsf{c}\,\gamma\,(\gamma^I\beta_I\mathsf{b})$ and $\partial\mathsf{c}\,\gamma\,(\beta_1\beta_2\beta_3)$. Their $SU(2)_R$-allowed target skeleton has dimension 8.

$$\ker(Q_0|_{\text{target skeleton}}) = 0 \quad [\text{p}], \qquad Q_0\delta\mathcal{O}_0 = -\delta Q_0\mathcal{O}_0 = 0 \quad \Longrightarrow \quad \delta\mathcal{O}_0 = 0.$$

□

The source colour space $\mathrm{Hom}_G(\mathbf{8}^{\otimes 3}, \Lambda^2\mathbf{8})$ has dimension 16 for $(\gamma^I\beta_I\mathsf{b})$. Cyclic $(\beta_1\beta_2\beta_3)$ and, for $h \neq 0$, $(\beta_I)^3$ contribute $6, 3$, giving at most 25 parameters. Without $SU(2)_R$, the three-dimensional Q_0-kernel reaches one independent target direction and can raise rank $1 \to 2$. For input edges $1o, 2o, 3o$ and $\lambda_2 = 0$, the Gui–Wang–Williams kernel is $-\frac{1}{24}(\lambda_1 \wedge \lambda_3)^2$ [33, Section 4.5], versus $+\frac{1}{24}(\lambda_P \wedge \lambda_Q)^2$ here. This holds for all twenty input-derivative patterns of order at most three. The one-loop sign is $+1$, and nilpotence fixes the relative convention.

S6.5 The vanishing locus

We track how the first two-loop ranks vary with the deformation parameters. Hesse symmetry and polynomial rank tests organize the vanishing curves and exceptional points listed below.

The symmetric cubic is $C = h_e(x^3 + y^3 + z^3) + 6\rho xyz$. Superpotentials related by GL_3 on γ^I give equivalent twisted theories. The order-24 Hesse group is generated by $(\rho, h_e) \mapsto (\rho, \omega h_e)$, $\omega^3 = 1$, and $(\rho, h_e) \mapsto -\frac{\sqrt{-3}}{3}(h_e - \rho, h_e + 2\rho)$. Its invariants are

$$\begin{aligned} S = \rho(h_e^3 - \rho^3), \qquad T = h_e^6 - 20h_e^3\rho^3 - 8\rho^6, \\ T^2 + 64S^3 = h_e^3(h_e^3 + 8\rho^3)^3, \\ \Delta_4 = h_e(h_e^3 + 8\rho^3) = \frac{16\,h(h^3 - (q-1)^3)}{(1+q)^4}. \end{aligned} \tag{S6.3}$$

C is singular exactly on $\Delta_4 = 0$, where the theory is GL_3-equivalent to a β-deformation [34, Section 3]. $C_8 = S^2 - 18S - T - 27$ factors over $\mathbb{Q}(\omega)$ into $q = 0, \infty$ and $1 - \omega^i q + \omega^j h = 0$, $i \neq 0$. Set $O_8 = 4S^2 - 36S - 3T$, irreducible over $\mathbb{Q}(\sqrt{-3})$.

$$\begin{aligned} \frac{(1+q)^8}{64} O_8 = (q^2 - 14q + 1)h^6 - 2(q-1)(q^4 - 7q^3 - 24q^2 - 7q + 1)h^3 \\ + (q-1)^4(q^2 + q + 1)^2. \end{aligned} \tag{S6.4}$$

Proposition S6.3.

Every block of the d_2 computation at the hits $E + J_L = 7, 8, \frac{17}{2}$ is affine in (ρ, h_e), and its ranks are constant on Hesse orbits. Modulo p, along random rational lines in the parameter plane, the curves on which $\operatorname{rank} d_2$ *drops from 1 to 0 are $\Delta_4 = 0$ at $E + J_L = 7$ and $\frac{17}{2}$, and $\Delta_4\, S\, O_8 = 0$ at $E + J_L = 8$. On $\Delta_4 = 0$, $d_2 = 0$ except at the eight points of the Hesse orbit of $(\rho, h_e) = (\pm\sqrt{-3}, 0)$, the β-deformations with $q^3 = 1$, where* $\operatorname{rank} d_2 = 1$ *at $E + J_L = 7$ and $\frac{17}{2}$. Near $\Delta_4 = 0$, d_2 vanishes to second order in the distance.*

Proof.

The blocks are fitted to the affine form at three points and checked at three further points; at the default point they reproduce the ranks of the census. Along a line, the rank-drop divisor of each block is the greatest common divisor of two randomly projected maximal minors, computed exactly modulo p by characteristic polynomials; the factors of Δ_4 are removed, and each remaining factor is reconstructed as a bivariate polynomial, checked on oblique lines and written in S and T. By (S1.18), $\operatorname{rank} d_2 = \operatorname{rank} M_2 - \operatorname{rank} M_1 - \operatorname{rank} M_2' + \operatorname{rank} M_1'$, and the orders of vanishing in h_e of the corresponding determinants are $7-5-4+4 = 8-6-5+5 = 11-9-8+8 = 2$ at the three hits. At $E + J_L = 7$, isolated drop points off these curves were excluded for every block except M_1 and M_2, by resultants of two projected minors; the failure probability of each step is of order $\deg / p$. □

Entries give source $\dim E_1$, outgoing $\operatorname{rank} d_1$, source and target $\dim E_2$, then $\operatorname{rank} d_2$.

point	$E+J_L=7$	$E+J_L=8$	$E+J_L=\frac{17}{2}$
generic	$5/3/2, 2/1$	$4/1/3, 3/1$	$5/3/2, 4/1$
β line, generic ρ	$5/2/3, 2/0$	$6/1/5, 3/0$	$5/2/3, 4/0$
triangle lines, Sklyanin point $(q,h)=(-\omega,1)$	$5/2/3, 2/0$	$6/1/5, 3/0$	$5/2/3, 4/0$
β line, $q^3=1$	$9/3/6, 1/1$	$14/6/8, 1/0$	$9/3/6, 3/1$
$S=0$	$5/3/2, 2/1$	$4/2/2, 2/0$	$5/3/2, 4/1$
$O_8=0$		$4/2/2, 2/0$	
$C_8=0$	$d_2=1$	$d_2=1$	$d_2=1$
$\mathcal{N}=4$ point	$47/0/47, 2/0$	$98/0/98, 5/0$	$129/0/129, 4/0$

On $SO_8=0$, the level-8 outgoing/incoming d_1 ranks rise $1,0\to 2,1$, and both E_2 dimensions fall $3\to 2$. The pure-h line lies on $S=0$. On $C_8=0$, the Sklyanin point scheme degenerates. The changes affect E_1 at $Y+1, Y+2$, with the same d_2.

S6.6 Fortuity of the sources

We compare the source cohomology at finite gauge rank with its stable value. The differences give lower bounds on the number of fortuitous classes and identify sectors in which every source is fortuitous.

For $N\geq Y+1$, $SU(N)$ computes stable cohomology, giving $f\geq h_{SU(3)}-h_{SU(N)}$ and $f=h_{SU(3)}$ when $h_{SU(N)}=0$ S1.5. Weyl-orbit summation reduces the $Y=7$ $SU(8)$ grid from 8.9×10^8 to 82,225 points.

source	$h_{SU(3)}$	large rank	h there	fortuitous classes
$P_{14}=(7,0;0,0)_5$	5	$SU(6)$	0	all 5
$(7,0;1,0)_5$	5	$SU(6)$	0	all 5
$(7,0;2,0)_5$	5	$SU(6)$	0	all 5
$(8,0;1,0)_6$	6	$SU(7)$	4	≥ 2
$P_{19}=(8,0;1,0)_7$	19	$SU(8)$	6	≥ 13
P_{21}, $c=0$: $(9,0;1,0)_7$	40	$SU(8)$	5	≥ 35
P_{21}, $c=1$: $(9,1;1,0)_7$	36	$SU(8)$	0	all 36

Classical dimensions h_{E_1} by gauge rank.

class	Y	$SU(3)$	$SU(4)$	$SU(5)$	$SU(6)$	$SU(7)$
$(7,0;0,0)$	5	5	0	0	0	0
$(7,0;0,0)$	6	0	16	0	0	0
$(7,0;0,0)$	7	0			11	23
$(1,0;2,2)$	4	2	2		2	
$(8,0;0,0)$	6	4	11	2		2
$(2,0;2,2)$	5				8	
$(7,0;1,0)$	5	5	0	0	0	
$(7,0;1,0)$	6	10	41	9	9	
$(1,0;3,2)$	4				4	

Blank entries were uncomputed. At $\mathcal{N} = 4$, sector $(7,0;0,0)$ has $h = 8, 47, 76, 42$ for $Y = 4, \ldots, 7$, all non-fortuitous. Deformation leaves five fortuitous classes at $Y = 5$, one lifted per triplet.

S6.7 The β-deformation

We repeat the census on the β-deformation line, where individual flavour weights remain available. The tables locate a rank-one two-loop lift and show how it arises from the first correction to the source.

For $h_e = 0$, $\rho = \frac{12345}{678}$, use the individual flavour weights of S1.1, with the flavour raising operators omitted. Complete census through $E + J_L = \frac{21}{2}$.

$E + J_L$	sectors	$\sum h_{E_1}$	d_1 pairs	$d_1 \neq 0$	d_2 pairs	$d_2 \neq 0$	d_3 pairs	$d_3 \neq 0$	$\sum h_{E_\infty}$
1	1	0	0	0	0	0	0	0	0
$\frac{3}{2}$	1	0	0	0	0	0	0	0	0
2	2	2	0	0	0	0	0	0	2
$\frac{5}{2}$	1	1	0	0	0	0	0	0	1
3	5	5	1	0	0	0	0	0	5
$\frac{7}{2}$	2	3	0	0	0	0	0	0	3
4	6	6	0	0	0	0	0	0	6
$\frac{9}{2}$	5	9	1	0	0	0	0	0	9
5	9	14	2	0	0	0	0	0	14
$\frac{11}{2}$	6	16	1	0	0	0	0	0	16
6	16	44	5	0	1	0	0	0	44
$\frac{13}{2}$	9	40	3	0	0	0	0	0	40
7	19	71	8	2	0	0	0	0	67
$\frac{15}{2}$	16	108	8	1	1	0	0	0	106
8	26	149	17	1	4	0	0	0	147
$\frac{17}{2}$	19	190	13	4	1	0	0	0	182
9	39	383	25	7	4	0	0	0	369
$\frac{19}{2}$	26	417	23	5	5	0	0	0	407
10	46	675	37	11	7	0	1	0	641
$\frac{21}{2}$	39	1054	38	13	8	1	1	0	1004

$E + J_L$	source	target	dim E_2 source	dim E_2 target	$Q_2\mathcal{K} \neq 0$	rank d_2
6	$(2,2,2;0,0)_5$	$(0,0,0;2,2)_4$	2	1	0	0
$\frac{15}{2}$	$(2,2,2;1,0)_5$	$(0,0,0;3,2)_4$	6	1	0	0
8	$(3,3,2;0,0)_6$	$(1,1,0;2,2)_5$	2	1	0	0
8	$(4,2,2;0,0)_5$	$(2,0,0;2,2)_4$	1	2	4	0
8	$(4,2,2;0,0)_6$	$(2,0,0;2,2)_5$	1	2	0	0
8	$(4,2,2;0,0)_7$	$(2,0,0;2,2)_6$	1	1	0	0
$\frac{17}{2}$	$(3,2,2;1,0)_6$	$(1,0,0;3,2)_5$	9	1	0	0
9	$(2,2,2;2,0)_5$	$(0,0,0;4,2)_4$	9	2	4330	0
9	$(3,3,3;0,0)_6$	$(1,1,1;2,2)_5$	2	2	2672	0
9	$(3,3,3;0,0)_7$	$(1,1,1;2,2)_6$	1	5	0	0

$E+J_L$	source	target	$\dim E_2$ source	$\dim E_2$ target	$Q_2\mathcal{K}\neq 0$	$\operatorname{rank} d_2$
9	$(5,2,2;0,0)_7$	$(3,0,0;2,2)_6$	3	2	0	0

$$\begin{gathered}(5,2,2;1,0)_8 \xrightarrow{d_2} (3,0,0;3,2)_7, \qquad \operatorname{rank} d_2 = 1 \text{ [p,p']},\\ (E;R_I;J_L,J_R):\ (\tfrac{19}{2};\tfrac{9}{2},\tfrac{3}{2},\tfrac{3}{2};1,\tfrac{1}{2}) \to (10;5,2,2;\tfrac{1}{2},\tfrac{1}{2}).\end{gathered} \tag{S6.5}$$

Here the source has $h=1$, the target has $\dim E_2 = 4$, and $Q_2\mathcal{O}_0 = 0$. Thus $d_2[\mathcal{O}_0]_2 = [Q_1\mathcal{O}_1]_2$.

S7 N=2 quiver, SQCD and mass deformations

We adapt the same charge-sector calculation to the $\mathcal{N}=2$ quiver, SQCD and their adjoint-mass deformations. The resulting tables record the successive differential ranks with the symmetries and gradings appropriate to each theory.

S7.1 The quiver and SQCD

We obtain the quiver complex by the orbifold projection and SQCD by decoupling one gauge node. A field rescaling relates nonzero coupling ratios, after which the census tests the surviving loop differentials.

The $SU(N)\times SU(N)$ quiver keeps $\mathsf{c},\mathsf{b},\gamma^3,\beta_3$ block-diagonal in $SU(2N)$ and $\gamma^{1,2},\beta_{1,2}$ off-diagonal. Its orbifold action is $C(\theta_1,\theta_2,\theta_3)\mapsto C(-\theta_1,-\theta_2,\theta_3)$. Use the grading of S1.1 and highest weights of $GL(2)_{12}\times GL(1)_3\times SL(2)$.

Proposition S7.1.

For $\kappa = g_2/g_1 \neq 0$, $E_r(\kappa)\cong E_r(1)$ for every r.

Proof.

The power of κ^2 counts consumed minus produced second-node b,β_3 letters. Rescaling these letters by κ^2 gives $Q_n(\kappa) = \phi_\kappa^{-1} Q_n(1)\phi_\kappa$ and conjugate transferred differentials. □

The $\kappa\to 0$ limit gives $\mathcal{N}=2$ SQCD with $N_f = 2N$. Compute flavour singlets using the second-node Haar measure. Quiver census at $\kappa^2 = 1, 9$, followed by SQCD. $Q_2\mathcal{K}\neq 0$ counts nonzero direct actions.

N	$E+J_L$	sectors	blocks	$\sum h$	$d_1\neq 0$	$\sum \dim E_2$	d_2 pairs	max rank d_2	$Q_2\mathcal{K}\neq 0$
2	2	4	3	4	0	4	0	–	–
2	$\frac{5}{2}$	2	1	2	0	2	0	–	–
2	3	8	4	6	0	6	0	–	–
2	$\frac{7}{2}$	4	4	6	0	6	0	–	–
2	4	13	11	17	0	17	0	–	–
2	$\frac{9}{2}$	8	7	14	0	14	0	–	–
2	5	20	19	38	1	36	0	–	–
2	$\frac{11}{2}$	13	15	42	1	40	0	–	–
2	6	31	34	93	2	89	0	–	–
2	$\frac{13}{2}$	20	26	116	1	112	0	–	–
2	7	44	57	240	3	230	1	0	0
2	$\frac{15}{2}$	31	43	276	3	268	1	0	0

N	$E + J_L$	sectors	blocks	$\sum h$	$d_1 \neq 0$	$\sum \dim E_2$	d_2 pairs	max rank d_2	$Q_2\mathcal{K} \neq 0$
2	8	61	90	560	5	544	3	0	4
2	$\frac{17}{2}$	44	73	716	4	696	1	0	2
2	9	84	140	1298	7	1258	7	0	10
2	$\frac{19}{2}$	61	110	1690	8	1640	5	0	8
2	10	111	205	3153	13	3059	11	0	18
2	$\frac{21}{2}$	84	169	4088	13	3960	8	0	16
3	2	4	3	4	0	4	0	–	–
3	$\frac{5}{2}$	2	1	2	0	2	0	–	–
3	3	8	5	8	0	8	0	–	–
3	$\frac{7}{2}$	4	5	8	0	8	0	–	–
3	4	13	12	21	0	21	0	–	–
3	$\frac{9}{2}$	8	8	18	0	18	0	–	–
3	5	20	24	54	0	54	0	–	–
3	$\frac{11}{2}$	13	19	62	0	62	0	–	–
3	6	31	40	149	1	147	1	0	0
3	$\frac{13}{2}$	20	32	176	1	174	0	–	–
3	7	44	69	400	4	390	2	0	0
3	$\frac{15}{2}$	31	54	478	3	468	1	0	0
3	8	61	113	1016	8	986	8	0	4
3	$\frac{17}{2}$	44	92	1304	7	1270	3	0	2
3	9	84	171	2634	19	2542	11	0	12

SQCD census.

N	$E + J_L$	sectors	blocks	$\sum h$	$d_1 \neq 0$	$\sum \dim E_2$	d_2 pairs	max rank d_2	$Q_2\mathcal{K} \neq 0$
2	2	4	3	3	0	3	0	–	–
2	$\frac{5}{2}$	2	1	1	0	1	0	–	–
2	3	8	3	4	0	4	0	–	–
2	$\frac{7}{2}$	4	3	4	0	4	0	–	–
2	4	13	7	10	0	10	0	–	–
2	$\frac{9}{2}$	8	4	6	0	6	0	–	–
2	5	20	9	21	0	21	0	–	–
2	$\frac{11}{2}$	13	9	21	0	21	0	–	–
2	6	31	19	37	0	37	0	–	–
2	$\frac{13}{2}$	20	14	54	0	54	0	–	–
2	7	44	26	112	0	112	0	–	–
2	$\frac{15}{2}$	31	23	109	0	109	0	–	–
2	8	61	42	232	0	232	1	0	0
2	$\frac{17}{2}$	44	30	307	0	307	1	0	1
2	9	84	59	481	0	481	1	0	1
2	$\frac{19}{2}$	61	54	623	0	623	1	0	1
2	10	111	87	1263	0	1263	4	0	4
2	$\frac{21}{2}$	84	75	1527	0	1527	5	0	5
2	11	144	122	2632	0	2632	8	0	8

N	$E+J_L$	sectors	blocks	$\sum h$	$d_1\neq 0$	$\sum \dim E_2$	d_2 pairs	max rank d_2	$Q_2\mathcal{K}\neq 0$
2	$\frac{23}{2}$	111	106	3850	0	3850	7	0	7
2	12	186	155	6440	0	6440	14	0	14
3	2	4	3	3	0	3	0	–	–
3	$\frac{5}{2}$	2	1	1	0	1	0	–	–
3	3	8	4	5	0	5	0	–	–
3	$\frac{7}{2}$	4	4	5	0	5	0	–	–
3	4	13	9	12	0	12	0	–	–
3	$\frac{9}{2}$	8	6	8	0	8	0	–	–
3	5	20	14	25	0	25	0	–	–
3	$\frac{11}{2}$	13	12	26	0	26	0	–	–
3	6	31	26	55	0	55	0	–	–
3	$\frac{13}{2}$	20	23	66	0	66	0	–	–
3	7	44	40	150	1	148	1	0	0
3	$\frac{15}{2}$	31	35	161	1	159	1	0	0
3	8	61	69	323	2	319	2	0	0
3	$\frac{17}{2}$	44	56	424	1	420	2	0	1
3	9	84	101	781	3	773	7	0	4

All quiver d_3 pairs at levels $9, 10$ have rank zero at both couplings. The control $Q_2 \mapsto 4Q_2$ gives rank 3 on the corresponding $SU(2)\times SU(2)$ pairs. For SQCD at $N=3$, nonzero d_1 occur in $(1,1,5;0,0)_6$, $(1,1,4;1,0)_6$, $(1,1,3;1,1)_6$, $(1,1,6;0,0)_6$, $(1,1,5;1,0)_6$, $(1,1,4;1,1)_6$, $(1,1,4;2,0)_6$, $(1,1,7;0,0)_8$. At $N=2$, $d_1=0$ through level 12.

S7.2 Mass deformations

We add the adjoint mass to the classical differential and reorganize the calculation by the surviving charges. The census then tests lifting in the quartic theories reached by the mass deformation.

An adjoint mass adds $\Lambda^{AB}(\gamma^3)_B\partial/\partial(\beta_3)^A$ to Q_0. Use the massless $Q_{n\geq 1}$. The flows give quartic S8.4, quartic SQCD, and Klebanov–Witten for masses $(M,-M)$. The mass breaks θ_3. Use the grading

$$\begin{gathered} A=\theta_1+\theta_3, \qquad B=\theta_2+\theta_3, \qquad \mathrm{gh}=Y-\Theta, \qquad R_{\mathrm{IR}}=\tfrac{1}{2}(R_1+R_2)+R_3, \\ E_{\mathrm{IR}}=\tfrac{3}{2}R_{\mathrm{IR}}+2J_L, \qquad w=A+B+2(n_++n_-)=\tfrac{4}{3}(E_{\mathrm{IR}}+J_L), \\ d_r:(A,B;n_\pm;\mathrm{gh})\to(A-2r,B-2r;n_\pm+r;\mathrm{gh}+1+2r), \end{gathered} \tag{S7.1}$$

The one-mass $SU(2)$ classes of [35].

class	$(A,B;n_+,n_-;\mathrm{gh})$	J_L	R_{IR}	E_{IR}	quantum
w	$(3,3;0,0;-2)$	1	1	$\frac{7}{2}$	graviton
w^2	$(6,6;0,0;-4)$	2	2	7	closed completion to $\hbar^3$ (Section S8.4)
$\mathcal{O}_7$	$(5,2;0,0;-2)$	1	$\frac{3}{2}$	$\frac{17}{4}$	$d_1=0$, $r\geq 2$ empty: all orders [p]
$\mathcal{O}_{16}$	$(8,8;0,0;-5)$	$\frac{5}{2}$	3	$\frac{19}{2}$	$d_1,d_2,d_3,d_4=0$, no d_5, no incoming: all orders [p]

For $\mathcal{O}_{16}$, the d_1,d_2,d_3 targets have $h=0,7,8$, with $Q_2\mathcal{O}_{16}\neq 0$ and $Q_4\mathcal{O}_{16}=0$.

theory	w	sectors	blocks	$\sum h$	$d_1 \neq 0$	$\sum \dim E_2$	d_r pairs	r tested
$\mathcal{N}$=4 $SU(2)$, one mass	4	7	4	4	0	4	–	–
	5	7	4	4	0	4	–	–
	6	13	11	11	0	11	–	–
	7	13	12	13	0	13	–	–
	8	22	25	30	0	30	–	–
	9	22	24	40	0	40	–	–
	10	34	51	81	0	81	–	–
	11	34	49	115	0	115	–	–
	12	50	93	218	0	218	4	d_2
	13	50	90	317	0	317	4	d_2
	14	70	154	569	0	569	11	d_2
	15	70	151	845	0	845	14	d_2, d_3
	16	95	240	1473	1	1471	32	d_2, d_3
	17	95	241	2194	0	2194	36	d_2, d_3
$\mathcal{N}$=4 $SU(3)$, one mass	4	7	6	6	0	6	–	–
	5	7	10	10	0	10	–	–
	6	13	19	23	0	23	–	–
	7	13	27	42	0	42	–	–
	8	22	50	96	0	96	1	d_2
	9	22	58	171	0	171	1	d_2
	10	34	98	358	0	358	5	d_2
	11	34	112	641	0	641	7	d_2
	12	50	178	1284	0	1284	15	d_2
	13	50	194	2300	0	2300	20	d_2, d_3
SCQCD $SU(2)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	16	46	0	46	–	–
	7	13	0	0	0	0	–	–
	8	22	34	194	0	194	–	–
	9	22	0	0	0	0	–	–
	10	34	62	826	0	826	2	d_2
	11	34	0	0	0	0	–	–
	12	50	109	3371	0	3371	7	d_2
	13	50	0	0	0	0	–	–
	14	70	173	13622	0	13622	18	d_2, d_3
	15	70	0	0	0	0	–	–
SCQCD $SU(3)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	18	44	0	44	–	–
	7	13	0	0	0	0	–	–
	8	22	35	203	0	203	1	d_2
	9	22	0	0	0	0	–	–
	10	34	71	934	0	934	3	d_2
	11	34	0	0	0	0	–	–
	12	50	125	4430	0	4430	9	d_2

theory	w	sectors	blocks	$\sum h$	$d_1 \neq 0$	$\sum \dim E_2$	d_r pairs	r tested
	13	50	0	0	0	0	–	–
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M,-M)$	4	7	10	16	0	16	–	–
	5	7	0	0	0	0	–	–
	6	13	26	66	0	66	–	–
	7	13	0	0	0	0	–	–
	8	22	53	299	1	297	–	–
	9	22	0	0	0	0	–	–
	10	34	98	1306	1	1304	3	d_2
	11	34	0	0	0	0	–	–
	12	50	170	5442	1	5440	13	d_2, d_3
	13	50	0	0	0	0	–	–

All d_2, d_3 ranks vanish. Nonzero d_1 occur at $(6,6;1,1;-3)$ in one-mass $SU(2)$ and $(2,2;k,1;1)$, $k=1,2,3$, in the quiver. At the massive-SQED weight S8.1, one-mass $SU(2)$ has $h=0$ and mass-deformed SQCD has $E_1=0$. The neighbouring $SU(2)$ ghost number has $h=1, d_2=0$.

S8 Explicit representatives and obstruction certificates

We now display representatives and their corrections so that the lifting tests can be checked directly. Massive SQED supplies a nonzero obstruction; the remaining examples give finite closed completions and tests of nonexactness.

S8.1 Massive SQED two-loop obstruction

We follow one massive-SQED class through its first correction and its second-order obstruction. Eliminating the massive pair reduces the final test to a rational rank calculation in the light-field complex.

Eliminate the massive pair and test the second-order residual.

Take $U(1)$ matter of charges $+1,-1,0$, with neutral mass m and cubic coupling y.

$$\mathcal{I} = ig\mathsf{c}(\gamma_+\beta^+ - \gamma_-\beta^-) + \frac{m}{2}\gamma_s^2 + y\gamma_s\gamma_+\gamma_-, \qquad my \neq 0, \tag{S8.1}$$

The massive pair. We separate the massive fields by a canonical change of variables. Contracting this pair gives the quartic light-field interaction and the transferred loop operators.

Eliminate the massive pair by the canonical shift

$$\gamma_s' = \gamma_s + \frac{y}{m}\gamma_+\gamma_-, \qquad \beta^{+\prime} = \beta^+ - \frac{y}{m}\gamma_-\beta^s, \qquad \beta^{-\prime} = \beta^- - \frac{y}{m}\gamma_+\beta^s \tag{S8.2}$$

The shift preserves the cotangent pairing and completes the square.

$$\frac{m}{2}\gamma_s^2 + y\gamma_s\gamma_+\gamma_- = \frac{m}{2}(\gamma_s')^2 - \frac{y^2}{2m}(\gamma_+\gamma_-)^2. \tag{S8.3}$$

The pair $Q_0\beta^s = -m\gamma_s'$, $Q_0\gamma_s' = 0$ is contractible.

$$\bar{W}_{\text{eff}} = -\frac{y^2}{2m}(\gamma_+\gamma_-)^2, \qquad \bar{W}_{\text{eff}}^{++--} = -\frac{2y^2}{m}. \tag{S8.4}$$

The superscript $++--$ takes two derivatives in each of γ_+, γ_-.

Express Q_n in the shifted fields, then transfer to the light fields.

Here π sets γ_s', β^s and their jets to zero, ι includes light fields, and h contracts the massive pair.

$$\begin{aligned} Q_0 h + h Q_0 &= 1 - \iota\pi, \\ Q_{1,\text{eff}} &= \pi Q_1 \iota, \\ Q_{2,\text{eff}} &= \pi Q_2 \iota - \pi Q_1 h Q_1 \iota. \end{aligned} \tag{S8.5}$$

The source. We choose a gauge-invariant operator and verify its classical closure term by term. Rewriting it in the shifted fields prepares the same class for the light-field calculation.

For $gmy \neq 0$, choose the gauge-invariant source below, with free dotted index $\dot{a}$.

$$\begin{aligned} \mathcal{O}_0 = ig\Big[&\beta^+\beta^-\mathsf{b}^2 + \frac{y}{m}(\gamma_-\beta^- - \gamma_+\beta^+)\beta^s\mathsf{b}^2 \\ &+ \frac{y^2}{3m\,ig}\gamma_+\gamma_-\mathsf{b}^3\Big]\partial_{+\dot{a}}\mathsf{c}. \end{aligned} \tag{S8.6}$$

Use $Q_0\mathsf{c} = 0$ and the following three variations to check $Q_0\mathcal{O}_0 = 0$.

$$\begin{aligned} Q_0(\beta^+\beta^-\mathsf{b}^2) &= y\gamma_s(\gamma_+\beta^+ - \gamma_-\beta^-)\mathsf{b}^2, \\ Q_0[(\gamma_-\beta^- - \gamma_+\beta^+)\beta^s\mathsf{b}^2] &= -(\gamma_+\beta^+ - \gamma_-\beta^-)(m\gamma_s + y\gamma_+\gamma_-)\mathsf{b}^2, \\ Q_0(\gamma_+\gamma_-\mathsf{b}^3) &= 3ig\gamma_+\gamma_-(\gamma_+\beta^+ - \gamma_-\beta^-)\mathsf{b}^2. \end{aligned} \tag{S8.7}$$

In the shifted fields, the same source is

$$\mathcal{O}_0 = ig\left(\beta^{+\prime}\beta^{-\prime}\mathsf{b}^2 - \frac{\bar{W}^{++--}_{\text{eff}}}{6ig}\gamma_+\gamma_-\mathsf{b}^3\right)\partial_{+\dot{a}}\mathsf{c}, \qquad \bar{W}^{++--}_{\text{eff}} = -\frac{2y^2}{m}. \tag{S8.8}$$

One loop. We evaluate the triangle on the chosen source and solve the first correction equation. The displayed correction cancels the one-loop variation and permits the two-loop obstruction test.

The triangle acts on this source as follows.

$$\begin{aligned} (64\pi^2)Q_1\mathcal{O}_0 = 2(ig)^3\Big\{&\mathsf{b}\,\partial_{+\dot{b}}(\beta^{+\prime}\beta^{-\prime})\partial_+{}^{\dot{b}}\mathsf{c} - (\partial_{+\dot{b}}\beta^{+\prime})\beta^{+\prime}\beta^{-\prime}\partial_+{}^{\dot{b}}\gamma_+ \\ &+ \beta^{+\prime}(\partial_{+\dot{b}}\beta^{-\prime})\beta^{-\prime}\partial_+{}^{\dot{b}}\gamma_- + \frac{y}{m}(\partial_{+\dot{b}}\beta^s)\mathsf{b}(\beta^{-\prime}\gamma_- - \beta^{+\prime}\gamma_+)\partial_+{}^{\dot{b}}\mathsf{c} \\ &- \frac{y}{m}(\partial_{+\dot{b}}\beta^s)\beta^{+\prime}\beta^{-\prime}\partial_+{}^{\dot{b}}(\gamma_+\gamma_-)\Big\}\partial_{+\dot{a}}\mathsf{c} \\ &+ 2(ig)^2y\Big\{\mathsf{b}^2(\partial_{+\dot{b}}\mathsf{c})\partial_+{}^{\dot{b}}\gamma_s' - \mathsf{b}(\partial_{+\dot{b}}\gamma_s')(\beta^{+\prime}\partial_+{}^{\dot{b}}\gamma_+ + \beta^{-\prime}\partial_+{}^{\dot{b}}\gamma_-)\Big\}\partial_{+\dot{a}}\mathsf{c} \\ &- \frac{2(ig)^2y^2}{m}\Big\{2\mathsf{b}^2(\partial_{+\dot{b}}\mathsf{c})\partial_+{}^{\dot{b}}(\gamma_+\gamma_-) + 3\mathsf{b}(\beta^{+\prime}\gamma_+ - \beta^{-\prime}\gamma_-)(\partial_{+\dot{b}}\gamma_+)\partial_+{}^{\dot{b}}\gamma_- \\ &+ \mathsf{b}\gamma_+\gamma_-[(\partial_{+\dot{b}}\beta^{+\prime})\partial_+{}^{\dot{b}}\gamma_+ + (\partial_{+\dot{b}}\beta^{-\prime})\partial_+{}^{\dot{b}}\gamma_-] \\ &+ \frac{y}{m}\mathsf{b}\gamma_+\gamma_-(\partial_{+\dot{b}}\beta^s)\partial_+{}^{\dot{b}}(\gamma_+\gamma_-)\Big\}\partial_{+\dot{a}}\mathsf{c}. \end{aligned} \tag{S8.9}$$

Solve $Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0$ with

$$
\begin{aligned}
(64\pi^2)\mathcal{O}_1 = \frac{2}{3}\Big[&(ig)^2\beta^{+\prime}\beta^{-\prime} + \frac{igy^2}{m}\mathsf{b}\gamma_+\gamma_-\Big] \\
&\times\Big[(\partial_{+\dot{a}}\partial_{+\dot{b}}\beta^{+\prime})\partial_+{}^{\dot{b}}\gamma_+ - (\partial_{+\dot{b}}\beta^{+\prime})\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_+ \\
&\quad - (\partial_{+\dot{a}}\partial_{+\dot{b}}\beta^{-\prime})\partial_+{}^{\dot{b}}\gamma_- + (\partial_{+\dot{b}}\beta^{-\prime})\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_-\Big] \\
&+ \frac{2(ig)^2}{3}\mathsf{b}\Big\{\big[(\partial_{+\dot{b}}\beta^{+\prime})\beta^{-\prime} - \beta^{+\prime}\partial_{+\dot{b}}\beta^{-\prime}\big]\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\mathsf{c} \\
&\quad - \big[(\partial_{+\dot{a}}\partial_{+\dot{b}}\beta^{+\prime})\beta^{-\prime} - \beta^{+\prime}\partial_{+\dot{a}}\partial_{+\dot{b}}\beta^{-\prime}\big]\partial_+{}^{\dot{b}}\mathsf{c}\Big\} \\
&+ \frac{8igy^2}{3m}\mathsf{b}(\beta^{+\prime}\partial_{+\dot{a}}\gamma_+ + \beta^{-\prime}\partial_{+\dot{a}}\gamma_-)(\partial_{+\dot{b}}\gamma_+)\partial_+{}^{\dot{b}}\gamma_- \\
&+ \frac{4igy^2}{3m}\mathsf{b}\Big[\beta^{+\prime}\gamma_-(\partial_{+\dot{b}}\gamma_+)\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_+ - \beta^{-\prime}\gamma_+(\partial_{+\dot{b}}\gamma_-)\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_-\Big] \\
&+ \frac{igy^2}{3m}\mathsf{b}^2\Big\{(\partial_{+\dot{b}}\mathsf{c})(\gamma_-\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_+ - \gamma_+\partial_{+\dot{a}}\partial_+{}^{\dot{b}}\gamma_-) \\
&\quad + (\partial_{+\dot{a}}\partial_{+\dot{b}}\mathsf{c})(\gamma_+\partial_+{}^{\dot{b}}\gamma_- - \gamma_-\partial_+{}^{\dot{b}}\gamma_+) + 15(\partial_{+\dot{a}}\mathsf{c})(\partial_{+\dot{b}}\gamma_+)\partial_+{}^{\dot{b}}\gamma_-\Big\} \\
&+ \frac{(ig)^2y}{m}\Big\{\mathsf{b}^2(\partial_{+\dot{a}}\beta^s)(\partial_{+\dot{b}}\mathsf{c})\partial_+{}^{\dot{b}}\mathsf{c} \\
&\quad - 2\mathsf{b}(\partial_{+\dot{b}}\beta^s)(\beta^{+\prime}\partial_+{}^{\dot{b}}\gamma_+ + \beta^{-\prime}\partial_+{}^{\dot{b}}\gamma_-)\partial_{+\dot{a}}\mathsf{c}\Big\}.
\end{aligned}
\tag{S8.10}
$$

$$
Q_0\mathcal{O}_1 = -Q_1\mathcal{O}_0. \tag{S8.11}
$$

The second obstruction. We combine the direct two-loop action with the one-loop action on the first correction. Field weights and derivative totals specify the finite cochain spaces needed to test this residual.

Test the combined second-order residual.

$$
d_2[\mathcal{O}_0]_2 = [Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1]_2. \tag{S8.12}
$$

Enumerate gauge-invariant light-field monomials by fermion number, field weight, and derivative totals in the two holomorphic directions.

Field	$U(1)$ charge	Fermion number	Field weight
γ_+	1	0	1
γ_-	-1	0	1
β^+	-1	1	3
β^-	1	1	3
γ_s	0	0	2
β^s	0	1	2
b	0	0	4
c	0	1	0

The parameters m, y, g and holomorphic derivatives have weight zero.

Each interaction and contracted pair has weight four, so Q_n changes field weight by $4(n+1) - 4(2n+1) = -4n$ and derivative totals by (n, n).

The homotopy preserves these degrees because γ_s' and β^s both have weight two.

Work in the relative gauge-invariant complex, with differentiated ghosts and gauge-invariant light-field polynomials.

The quotient test. We test the residual after quotienting by classical boundaries and changes of the first correction. A rank increase and a rational functional give independent certificates that the obstruction is nonzero.

For $\dot{a} = \dot{+}$, rational elimination gives

Space or map	Dimension or rank
Odd source space, weight 14, derivatives $(1, 0)$	33
Classical source cycles and boundaries	18, 17
First-correction space, weight 10, derivatives $(2, 1)$	365
Classical cycles in the first-correction space	180
Second primitives, weight 6, derivatives $(3, 2)$	752
Columns of $\operatorname{im} Q_0 + Q_{1,\mathrm{eff}} \ker Q_0$	932
Rank of these columns	394
Rank after adjoining the second obstruction	395

The source defines one surviving class since $18 - 17 = 1$ and a d_1-preimage would require derivative totals $(0, -1)$.

For second-order candidates V and first-correction cycles Z, the quotient removes $Q_0 V$ and $Q_{1,\mathrm{eff}} Z$, with $Q_0 Z = 0$.

The exact coefficient archive supplies a rational functional ℓ annihilating both freedoms.

$$\begin{aligned} \ell(Q_0 V) = 0, \qquad \ell(Q_{1,\mathrm{eff}} Z) = 0 \quad (Q_0 Z = 0), \\ \ell\pi(Q_2 \mathcal{O}_0 + Q_1 \mathcal{O}_1) = -\frac{16(ig)^2 y^4}{m^2 (64\pi^2)^2}. \end{aligned} \tag{S8.13}$$

The residual class is

$$\begin{aligned} d_2[\mathcal{O}_0]_2 = -\frac{2(ig)^2 y^4}{m^2 (64\pi^2)^2} \Big[\partial_{+\dot{a}} (\gamma_+ \gamma_-) \\ \times \partial_{+\dot{b}} \partial_{+\dot{c}} (\gamma_+ \gamma_-) \partial_+{}^{\dot{b}} \partial_+{}^{\dot{c}} (\gamma_+ \gamma_-) \Big]_2 \neq 0. \end{aligned} \tag{S8.14}$$

The rank increase $394 \to 395$ and the nonzero value in (S8.13) certify the two-loop lift.

S8.2 Complete multi-graviton checks in $SU(2)$ and $SU(3)$

We enumerate complete finite sectors of multi-graviton classes for $SU(2)$ and $SU(3)$. Explicit corrections verify closure, while Cartan restriction and the derivative bounds certify survival.

Compute the finite-sector completions.

Quantum multiplication completes these multi-graviton classes by Theorem 8.1.

Use the two-letter single graviton [7, Eq. (2.14)]

$$(w_2)^I = \mathrm{Tr}\left(\frac{\gamma^I \mathsf{b}}{ig} - \frac{1}{2}\varepsilon^{IJK}\beta_J\beta_K\right). \tag{S8.15}$$

For $SU(2)$, enumerate zero-jet invariants with at most four letters and $\deg_\theta - \#\text{letters} = 3$.

Flavour weights count superfield θ_I indices, with γ^I contributing I, β_I the complementary pair, and b all three.

The nonzero classical cohomology sectors have four letters.

flavour weight	invariants	$\dim\ker Q_0$	$\dim\mathrm{im}\, Q_0$	$\dim H(Q_0)$	orbit size
$(3,2,2)$	13	4	1	3	3
$(3,3,1)$	5	2	1	1	3
$(4,2,1)$	3	1	0	1	6

(S8.16)

Flavour permutations give $3\cdot 3 + 1\cdot 3 + 1\cdot 6 = 18$ classes.

Cartan restriction kills Q_0 boundaries and has ranks $3,1,1$ on these blocks, certifying their independence.

Generate representatives from the five seeds by flavour permutations preserving ε^{IJK}.

$$\begin{aligned}
\mathcal{O}_0^{(1)} &= -\,\mathrm{Tr}(\beta_2\gamma^1)(w_2)^2,\\
\mathcal{O}_0^{(2)} &= -\big[\mathrm{Tr}(\beta_2\gamma^2) - \mathrm{Tr}(\beta_1\gamma^1)\big](w_2)^1,\\
\mathcal{O}_0^{(3)} &= -\,\mathrm{Tr}(\beta_3\gamma^1)(w_2)^3,\\
\mathcal{O}_0^{(4)} &= -\,\mathrm{Tr}(\beta_3\gamma^1)(w_2)^2,\\
\mathcal{O}_0^{(5)} &= -\,\mathrm{Tr}(\beta_3\gamma^1)(w_2)^1.
\end{aligned} \tag{S8.17}$$

For the fifth seed, the first correction is

$$\begin{aligned}
\mathcal{O}_1^{(5)} = -\frac{i}{64\pi^2}C_A{}^{BC}\Big[&(\beta_3)^A(\partial_{+\dot a}\gamma^1)_B(\partial_+{}^{\dot a}\gamma^1)_C\\
&- 2(\partial_{+\dot a}\beta_3)^A(\gamma^1)_B(\partial_+{}^{\dot a}\gamma^1)_C\Big].
\end{aligned} \tag{S8.18}$$

All five seeds satisfy

$$\begin{gathered}
Q_1\mathcal{O}_0^{(j)} = -Q_0\mathcal{O}_1^{(j)} \neq 0, \qquad Q_2\mathcal{O}_0^{(j)} = Q_1\mathcal{O}_1^{(j)} = 0,\\
\mathbf{Q}(\mathcal{O}_0^{(j)} + \hbar\mathcal{O}_1^{(j)}) = 0.
\end{gathered} \tag{S8.19}$$

The coefficient degrees are $7, 4$, and closure equations at order $r \geq 3$ have degree $7 - 3r < 0$.

Corollary S1.2 excludes incoming differentials for these zero-jet classes.

For $SU(3)$, enumerate five-letter zero-jet invariants of flavour weight $(3,3,2)$.

$$\dim(\text{sources}) = 147, \qquad \dim\ker Q_0 = 35, \qquad \dim\mathrm{im}\, Q_0 = 26, \qquad \dim H(Q_0) = 9. \tag{S8.20}$$

The nine classes are products of two- and three-letter graviton traces, with Cartan-restriction rank nine.

One representative and its corrections are

$$\mathcal{O}_0 = \mathrm{Tr}\left(\frac{\mathsf{b}\gamma^2}{ig} + \beta_1\beta_3\right)\mathrm{Tr}(\beta_2\gamma^1\gamma^2 - \beta_3\gamma^3\gamma^1). \tag{S8.21}$$

Here $[X,Y]=XY-YX$ and $\{X,Y\}=XY+YX$, in the displayed field order.

$$\begin{aligned}\mathcal{O}_1=-\frac{1}{64\pi^2}\,\mathrm{Tr}\,\Big\{&(\partial_{+\dot a}\mathsf{c})[\partial_+{}^{\dot a}\mathsf{b},\gamma^1\gamma^2]\\&+ig(\partial_{+\dot a}\mathsf{c})(\{\gamma^1\beta_3,\partial_+{}^{\dot a}\beta_1\}+\{\gamma^2\beta_2-\beta_3\gamma^3,\partial_+{}^{\dot a}\beta_3\})\\&+\gamma^1\gamma^2([\partial_{+\dot a}\gamma^1,\partial_+{}^{\dot a}\beta_1]-[\partial_{+\dot a}\gamma^2,\partial_+{}^{\dot a}\beta_2]+[\partial_{+\dot a}\gamma^3,\partial_+{}^{\dot a}\beta_3])\\&+2\gamma^1(\partial_{+\dot a}\gamma^2)(-(\partial_+{}^{\dot a}\gamma^2)\beta_2+(\partial_+{}^{\dot a}\beta_3)\gamma^3)\\&+\gamma^1\beta_3[\partial_{+\dot a}\gamma^2,\partial_+{}^{\dot a}\gamma^3]+2\gamma^1(\partial_{+\dot a}\beta_3)(\partial_+{}^{\dot a}\gamma^2)\gamma^3\\&-(\partial_{+\dot a}\gamma^1)[\partial_+{}^{\dot a}\gamma^2,\gamma^2\beta_2-\beta_3\gamma^3]\Big\},\end{aligned}\tag{S8.22}$$

$$\begin{aligned}\mathcal{O}_2=-\frac{ig}{2(64\pi^2)^2}\,\mathrm{Tr}\,\Big[&(\partial_{+\dot a}\partial_{+\dot c}\mathsf{c})([\partial_+{}^{\dot a}\partial_+{}^{\dot c}\gamma^1,\gamma^2]\\&-ig(\partial_+{}^{\dot a}\partial_+{}^{\dot c}\mathsf{c})\beta_3)\Big].\end{aligned}\tag{S8.23}$$

Check the three equations

$$Q_1\mathcal{O}_0=-Q_0\mathcal{O}_1,\qquad Q_2\mathcal{O}_0=0,\qquad Q_1\mathcal{O}_1=-Q_0\mathcal{O}_2\neq 0.\tag{S8.24}$$

The coefficient degrees $8,5,2$ give closure degree $8-3r<0$ for $r\geq 3$.

Cartan independence and Corollary S1.2 certify nine surviving classes.

S8.3 The $SU(2)$ threshold class and its graviton hair

We complete the $SU(2)$ threshold class and then attach its graviton dressings by quantum multiplication. The final tests exclude incoming differentials for the three dressed classes, including the derivative hair.

Complete the threshold core and its graviton dressings.

The threshold class. We solve the correction equations for the lowest-charge $SU(2)$ fortuitous class. The coefficient degrees terminate the completion at fourth order, giving a closed perturbative representative.

Take the lowest-charge $SU(2)$ fortuitous class [7, Section 3.1].

$$\mathcal{O}_0=-\varepsilon^{IJK}\,\mathrm{Tr}(\gamma^L\beta_I)\,\mathrm{Tr}(\gamma^M\beta_J)\,\mathrm{Tr}(\beta_L\beta_M\beta_K).\tag{S8.25}$$

This $SU(3)_R$ singlet has two scalars, five fermions, and component degree twelve.

Solve its four representative equations.

$$\mathcal{O}[\mathcal{O}_0]=\sum_{r=0}^{4}\hbar^r\mathcal{O}_r,\qquad \sum_{n=0}^{r}Q_n\mathcal{O}_{r-n}=0\quad(0\leq r\leq 4).\tag{S8.26}$$

r	0	1	2	3	4
number of letters	7	6	5	4	3
$\deg_\theta$	12	9	6	3	0
number of derivatives	0	2	4	6	8

(S8.27)

The last coefficient is

$$\begin{aligned}\mathcal{O}_4 = &-\frac{42731(ig)^3}{432(32\pi^2)^4}\,\mathrm{Tr}\Big[(\partial_{+\dot a}\partial_{+\dot b}\partial_{+\dot c}\partial_{+\dot d}\mathsf{c})(\partial_+{}^{\dot a}\partial_+{}^{\dot b}\partial_+{}^{\dot c}\mathsf{c})(\partial_+{}^{\dot d}\mathsf{c})\Big]\\ &+\frac{23323(ig)^3}{216(32\pi^2)^4}\,\mathrm{Tr}\Big[(\partial_{+\dot a}\partial_{+\dot b}\partial_{+\dot c}\partial_{+\dot d}\mathsf{c})(\partial_+{}^{\dot a}\partial_+{}^{\dot b}\mathsf{c})(\partial_+{}^{\dot c}\partial_+{}^{\dot d}\mathsf{c})\Big]\\ &+\frac{90043(ig)^3}{432(32\pi^2)^4}\,\mathrm{Tr}\Big[(\partial_{+\dot a}\partial_{+\dot b}\partial_{+\dot c}\mathsf{c})(\partial_+{}^{\dot a}\partial_+{}^{\dot b}\partial_{+\dot d}\mathsf{c})(\partial_+{}^{\dot c}\partial_+{}^{\dot d}\mathsf{c})\Big].\end{aligned} \tag{S8.28}$$

The coefficient archive supplies $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$.

At two loops the cancellation is

$$Q_2\mathcal{O}_0 \neq 0, \qquad Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1 = -Q_0\mathcal{O}_2, \qquad d_2[\mathcal{O}_0]_2 = 0. \tag{S8.29}$$

The closure degree $12 - 3r < 0$ for $r \geq 5$ terminates the recursion.

$$\mathbf{Q}\mathcal{O}[\mathcal{O}_0] = 0. \tag{S8.30}$$

Classical nonexactness [7, Section 3.1] and Corollary S1.2 certify perturbative survival.

Graviton hair. We attach the two-letter graviton to the completed threshold class using the quantum product. The component supersymmetry transformations then give the completion of its second classical dressing.

Dress the core with $(w_2)^I$ of (S8.15) [7, Eq. (3.3)].

Translation of θ_I acts by

$$igQ_+^I\mathsf{c} = \gamma^I, \qquad Q_+^I\gamma^J = \varepsilon^{IJK}\beta_K, \qquad Q_+^I\beta_J = -\frac{\delta_J^I}{ig}\mathsf{b}, \qquad Q_+^I\mathsf{b} = 0. \tag{S8.31}$$

Theorem 8.1, common θ translation, and these component transformations give

$$Q_n(w_2)^I = 0, \qquad \{Q_n, Q_+^I\} = 0, \qquad Q_+^I(w_2)^K = 0. \tag{S8.32}$$

The primes and subscript in $\mathcal{O}_1', \mathcal{O}_1''$ label the classical hairy classes of [7, Section 3.2].

Complete the two classical products [7, Eqs. (3.3)–(3.4)] by quantum multiplication.

$$\begin{aligned}\mathcal{O}[(w_2)^I\mathcal{O}_0] &= (w_2)^I \star \mathcal{O}[\mathcal{O}_0],\\ \mathcal{O}_1' &= \varepsilon_{IJK}(Q_+^IQ_+^J\mathcal{O}_0)(w_2)^K,\\ \mathcal{O}[\mathcal{O}_1'] &= \varepsilon_{IJK}Q_+^IQ_+^J((w_2)^K \star \mathcal{O}[\mathcal{O}_0]).\end{aligned} \tag{S8.33}$$

The derivative hair $\mathcal{O}_1''$. We express the derivative dressing as a two-letter graviton factor multiplying the threshold class. Closure of this factor and the binary chain identity turn the classical product into a finite quantum completion.

For the derivative hair, differentiate the graviton current [7, Eq. (2.14)].

$$\begin{aligned}&\frac{1}{8}\partial_+{}^{\dot a}\,\mathrm{Tr}\Big[\mathsf{b}\partial_{+\dot a}\mathsf{c} + \frac{2}{3}(\partial_{+\dot a}\gamma^I)\beta_I - \frac{1}{3}\gamma^I\partial_{+\dot a}\beta_I\Big]\\ &\qquad = \frac{1}{8}\,\mathrm{Tr}\Big[(\partial_+{}^{\dot a}\mathsf{b})(\partial_{+\dot a}\mathsf{c}) - (\partial_+{}^{\dot a}\gamma^I)(\partial_{+\dot a}\beta_I)\Big].\end{aligned} \tag{S8.34}$$

Antisymmetry kills two derivatives on one field, while the remaining coefficients combine as $-2/3 - 1/3 = -1$.

The resulting nine-letter flavour singlet has derivative totals $(1,1)$ [7, Eq. (3.5)].

$$\mathcal{O}_1'' = \frac{1}{8}\,\mathrm{Tr}\Big[(\partial_+{}^{\dot a}\mathsf{b})(\partial_{+\dot a}\mathsf{c}) - (\partial_+{}^{\dot a}\gamma^I)(\partial_{+\dot a}\beta_I)\Big]\,\mathcal{O}_0. \tag{S8.35}$$

Its two-letter graviton factor equals

$$\begin{aligned}\frac{1}{8}\,\mathrm{Tr}\Big[(\partial_+{}^{\dot a}\mathsf{b})(\partial_{+\dot a}\mathsf{c}) - (\partial_+{}^{\dot a}\gamma^I)(\partial_{+\dot a}\beta_I)\Big]\\ = \frac{1}{16}\int d^3\theta\;\mathrm{Tr}\Big[(\partial_+{}^{\dot a}C)(\partial_{+\dot a}C)\Big]\,.\end{aligned} \tag{S8.36}$$

The classical variation is a trace commutator, the one-loop odd kernel vanishes at equal θ, and higher fans require at least three letters.

The binary chain identity (5.13) therefore gives

$$\begin{aligned}\mathcal{O}[\mathcal{O}_1''] = \frac{1}{8}\,\mathrm{Tr}\Big[(\partial_+{}^{\dot a}\mathsf{b})(\partial_{+\dot a}\mathsf{c}) - (\partial_+{}^{\dot a}\gamma^I)(\partial_{+\dot a}\beta_I)\Big]\star\mathcal{O}[\mathcal{O}_0],\\ \mathbf{Q}\mathcal{O}[\mathcal{O}_1''] = 0.\end{aligned} \tag{S8.37}$$

Coefficient degree $15-3r$ ends the completion at order five.

Exactness. We test whether the closed dressed representatives can become exact at a later page. Derivative bounds exclude the higher incoming maps, and the remaining one-loop possibility for the derivative hair is ruled out by the displayed image calculation.

For the first two zero-jet hairs, classical nonexactness [7, Section 3.2] and Corollary S1.2 exclude incoming differentials.

For the derivative hair, totals $(1,1)$ require any preimage to satisfy

$$\mathcal{O}_1'' = Q_1\Lambda_0 + Q_0\Lambda_1, \qquad Q_0\Lambda_0 = 0, \tag{S8.38}$$

Here Λ_0, Λ_1 are the leading preimage and its first correction.

Taking Q_0, Q_1 on the full Λ_1, Λ_0 candidate spaces with the required charges and flavour-singlet representation gives [7, Section 3.2]

$$\begin{gathered}\dim\mathrm{im}\,Q_0 = 912, \qquad \dim\mathrm{im}\,Q_1 = 25,\\ \dim(\mathrm{im}\,Q_0\cap\mathrm{im}\,Q_1) = 7,\\ \dim(\mathrm{im}\,Q_0+\mathrm{im}\,Q_1) = 912+25-7 = 930,\\ \mathcal{O}_1''\notin\mathrm{im}\,Q_0+\mathrm{im}\,Q_1.\end{gathered}$$

Corollary S1.2 excludes d_r-preimages for $r\geq 2$, completing the survival test for all three hairs.

S8.4 Explicit completions in the quartic deformation

We construct explicit completions of two fortuitous representatives in the quartic $SU(2)$ deformation. Spider operations give their corrections, and the field weights terminate the closure equations after finitely many orders.

Complete the two quartic representatives.

Classically eliminate one adjoint of mass M from $\mathcal{N}=4$ theory and take the resulting $SU(2)$ theory with constant gauge kinetic function.

$$\bar{W} = -\frac{1}{2M}\,\mathrm{Tr}[\gamma^1,\gamma^2]^2, \qquad M \neq 0. \tag{S8.39}$$

Apply the spider kernels of Section 6 to the quartic interaction.

First take the graviton square [35, Eq. (4.51)].

$$w = \frac{1}{igM}\,\mathrm{Tr}([\gamma^1,\gamma^2]\mathsf{b}) + \mathrm{Tr}(\beta_1\beta_2), \qquad \mathcal{O}_0 = w^2. \tag{S8.40}$$

On diagonal $SU(2)$ fields, $w^2 = 0$ by odd nilpotency.

For gauge-invariant polynomials, each surviving Q_0 variation introduces a scalar, so the scalar-free β^4 term proves classical nonexactness.

The spider operations give $\mathcal{O}[w^2] = w^2 + \hbar\mathcal{O}_1 + \hbar^2\mathcal{O}_2 + \hbar^3\mathcal{O}_3$.

The coefficient archive supplies the first two corrections, and the last is

$$\mathcal{O}_3 = -\frac{29g^2}{3M(64\pi^2)^3}\,\mathrm{Tr}\Big[(\partial_{+\dot{a}}\partial_{+\dot{b}}\partial_{+\dot{c}}\mathsf{c})(\partial_+{}^{\dot{a}}\partial_+{}^{\dot{b}}\partial_+{}^{\dot{c}}\mathsf{c})\Big]. \tag{S8.41}$$

At third order, check

$$\begin{aligned}
Q_3\mathcal{O}_0 &= -\frac{16g^3C_{ABC}}{M(64\pi^2)^3}(\partial_{+\dot{a}}\mathsf{c}^B)(\partial_{+\dot{b}}\partial_{+\dot{c}}\mathsf{c}^C)(\partial_+{}^{\dot{a}}\partial_+{}^{\dot{b}}\partial_+{}^{\dot{c}}\mathsf{c}^A),\\
Q_1\mathcal{O}_2 &= -\frac{42g^3C_{ABC}}{M(64\pi^2)^3}(\partial_{+\dot{a}}\mathsf{c}^B)(\partial_{+\dot{b}}\partial_{+\dot{c}}\mathsf{c}^C)(\partial_+{}^{\dot{a}}\partial_+{}^{\dot{b}}\partial_+{}^{\dot{c}}\mathsf{c}^A),\\
Q_0\mathcal{O}_3 &= \frac{58g^3C_{ABC}}{M(64\pi^2)^3}(\partial_{+\dot{a}}\mathsf{c}^B)(\partial_{+\dot{b}}\partial_{+\dot{c}}\mathsf{c}^C)(\partial_+{}^{\dot{a}}\partial_+{}^{\dot{b}}\partial_+{}^{\dot{c}}\mathsf{c}^A),\\
Q_2\mathcal{O}_1 &= 0.
\end{aligned} \tag{S8.42}$$

$$-16 - 42 + 58 = 0.$$

Weights $(1,3,4,0)$ for $(\gamma,\beta,\mathsf{b},\mathsf{c})$ give weight four to every interaction and contracted pair.

$$12 - 4r < 0 \quad (r \geq 4), \qquad \mathbf{Q}\mathcal{O}[w^2] = 0.$$

Classical nonexactness and Corollary S1.2 certify survival.

Next take the weight-sixteen flavour singlet $\mathcal{O}_{16}$ [35, Eq. (4.53)].

Use the source flavour indices $a,b,c = 1,2$, with $\epsilon^{12} = 1 = -\epsilon_{12}$, $\beta^a = \epsilon^{ab}\beta_b$, and $\gamma_a = \epsilon_{ab}\gamma^b$.

$$\begin{aligned}
\mathcal{O}_{16} = &-\frac{1}{6}C_{ABC}C_{DE}{}^{F}(\beta^a)^A(\beta^b)^B(\beta^c)^C(\beta_a)^D(\beta_b)^E(\gamma_c)_F\\
&+\frac{1}{(ig)^2M^2}C^{ABC}C_D{}^{EF}\delta^{GH}\mathsf{b}_A(\gamma^a)_B(\gamma_a)_C\mathsf{b}_G(\gamma^b)_H(\beta_b)^D(\gamma^c)_E(\gamma_c)_F\\
&-\frac{1}{gM}C_A{}^{BC}\delta^{DE}\delta_{FG}\mathsf{b}_D(\gamma^a)_E(\beta_a)^A(\gamma^b)_B(\gamma_b)_C(\beta^c)^F(\beta_c)^G\\
&-\frac{2}{gM}C_A{}^{BC}\mathsf{b}_D(\beta^a)^D(\gamma_a)_E(\beta^b)^E(\beta_b)^A(\gamma^c)_B(\gamma_c)_C.
\end{aligned} \tag{S8.43}$$

A preimage of the leading $\gamma\beta^5$ term would require a $\beta^4\mathsf{b}$ flavour singlet, absent among the $SU(2)$ contractions [35, Section 4.4].

The representative vanishes on diagonal fields.

For $M \neq 0$, solve

$$\mathcal{O}[\mathcal{O}_{16}] = \sum_{r=0}^{4} \hbar^r \mathcal{O}_r[\mathcal{O}_{16}], \qquad \sum_{n=0}^{r} Q_n \mathcal{O}_{r-n}[\mathcal{O}_{16}] = 0 \quad (0 \leq r \leq 4). \tag{S8.44}$$

The same archive supplies the four corrections, of weights $16 - 4r$, and their polynomial closure identities.

A four-loop operation contracts at most five input letters in (S8.43), of total weight at most fifteen, and lowers weight by sixteen.

$$Q_4 \mathcal{O}_{16} = 0, \qquad 16 - 4r < 0 \quad (r \geq 5), \qquad \mathbf{Q}\mathcal{O}[\mathcal{O}_{16}] = 0.$$

Classical nonexactness and Corollary S1.2 certify the resulting perturbative class.

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